PHYSICS

1) The greatest value of

$y = 2\sin\omega t + 2\cos\omega t + 4\sqrt{2}\sin(\omega t + \frac{\pi}{4})$

(1) $4 + 4\sqrt{2}$
(2) $4\sqrt{2}$
(3) $6\sqrt{2}$
(4) $10\sqrt{2}$

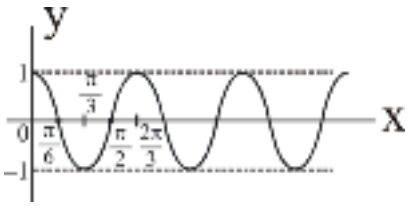
2) The value of integral $I = \int_1^2 \frac{dx}{3x+4}$ is :-

(1) $\frac{1}{3}\log_e\left(\frac{10}{7}\right)$
(2) $\log_e\left(\frac{10}{7}\right)$
(3) $\log_e\left(\frac{3}{4}\right)$
(4) $\log_e(2)$

3) Sun rays cast $4\sqrt{3}$ m long shadow of a pole when sun is 30° above the horizontal. When sun rises to 60° above the horizontal. Length of shadow becomes :-

(1) $\frac{4}{\sqrt{3}}$m
(2) $2\sqrt{3}$ m
(3) 2m
(4) $2\sqrt{2}$ m

4) The following graph corresponds to which equation:-



(1) y = cosx
(2) y = 3cosx
(3) y = 3cos3x
(4) y = cos3x

5) The moon is observed from two diametrically opposite points A & B on earth. The angle substended at the moon by two direction of observation is 1°30'. Given diameter of earth $\frac{720}{\pi} \times 10^6$ m. Then distance of moon from earth.

(1) 3.46×10^6 m

(2) 8.64×10^6 m

(3) 3.46×10^9 m

(4) 8.64×10^9 m

6) The value of the constant p such that the vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}-3\hat{k}$ and $3\hat{i}+p\hat{j}+5\hat{k}$ are coplanar is :

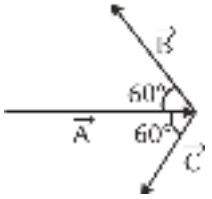
(1) zero

(2) 2

(3) –4

(4) –2

7) Three coplanar vectors $\vec{A}, \vec{B}$ and $\vec{C}$ have magnitudes 4, 3 and 2 respectively. If the angle between any two vectors is 120° then which of the following vector may be equal to $\frac{3\vec{A}}{4}+\frac{\vec{B}}{3}+\frac{\vec{C}}{2}$



(1) 

(2) 

(3) 

(4) 

8) Match the statements given in column-I with statements given in column-II

	Column-I		Column-II
A	if $\lvert\vec{A}\rvert = \lvert\vec{B}\rvert$ and $\lvert\vec{A}+\vec{B}\rvert = \lvert\vec{A}\rvert$ then angle between $\vec{A}$ and $\vec{B}$ is	p	90°
B	Magnitude of resultant of two forces $\lvert\vec{F}_1\rvert = 8N$ and $\lvert\vec{F}_2\rvert = 4N$ may be	q	120°
C	Angle between $\vec{A} = 2\hat{i}+2\hat{j}$ & $\vec{B} = 3\hat{k}$ is	r	12N

D	Magnitude of resultant of vectors $\vec{A} = 2\hat{i} + \hat{j}$ & $\vec{B} = 3\hat{k}$ is	s	$\sqrt{14}$

(1) (A) → q, (B) → r, (C) → p, (D) → s

(2) (A) → q, (B) → s, (C) → r, (D) → p

(3) (A) → s, (B) → r, (C) → p, (D) → q

(4) (A) → s, (B) → p, (C) → r, (D) → q

9) If $\vec{A} \times \vec{B} = \vec{C} + \vec{D}$, then select the correct alternative :-

(1) $\vec{B}$ is parallel to $\vec{C} + \vec{D}$

(2) $\vec{A}$ is perpendicular to $\vec{C}$

(3) Component of $\vec{C}$ along $\vec{A}$ = component of $\vec{D}$ along $\vec{A}$

(4) Component of $\vec{C}$ along $\vec{A}$ = –component of $\vec{D}$ along $\vec{A}$

10) If $\left|\vec{a} \times \vec{b}\right| = \frac{1}{\sqrt{3}}\left(\vec{a} \cdot \vec{b}\right)$, then $\left|\vec{a} + \vec{b}\right|$ is :-

(1) $\sqrt{a^2 + b^2 + ab}$

(2) $\sqrt{a^2 + b^2 + 2ab}$

(3) $\sqrt{a^2 + b^2}$

(4) $\sqrt{a^2 + b^2 + \sqrt{3}ab}$

11) Two particles are fired from the same point, each with speed 100 m/s, and firing angles with horizontal = 60° and 120° respectively. The time after which their velocity vectors become perpendicular to each other, is :

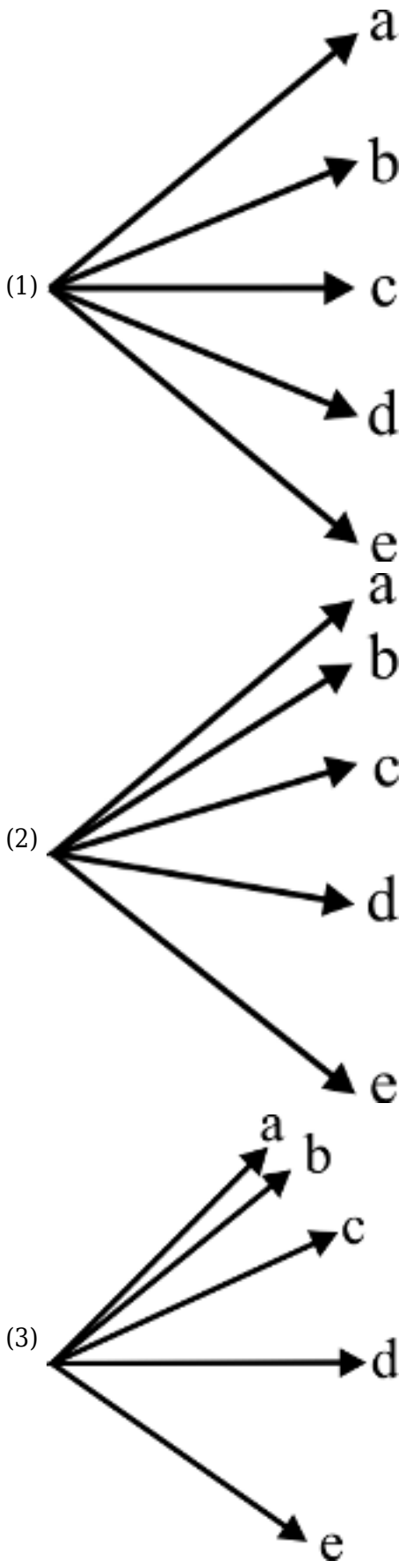
(1) 5 s

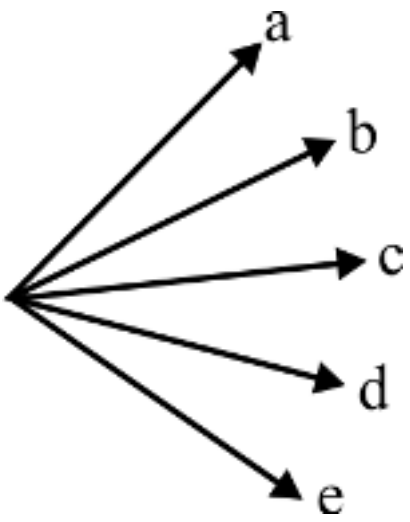
(2) $5\left(\sqrt{3}-1\right)s$

(3) $5\sqrt{3}s$

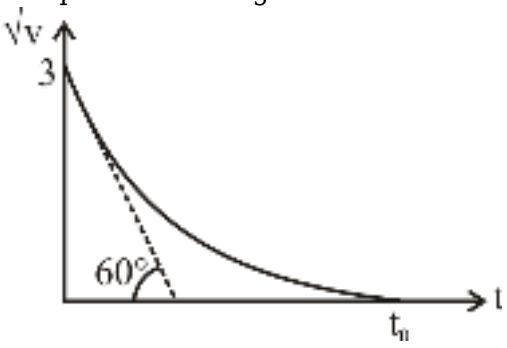
(4) $5\sqrt{3}/2s$

12) The velocity vectors of an object performing ground to ground projectile motion are drawn from starting time instant "a" to ending time "e" at fixed time intervals. Which of the following gives a possible vector diagram?


(1)
a
b
c
d
e
(2)
a
b
c
d
e
(3)
a
b
c
d
e

(4) 

13) Particle is moving along the x-axis with variable acceleration. Variation of square root of speed of a particle moving on the x-axis is as shown. Its initial magnitude of acceleration is :



(1) $2\sqrt{3}$ m/s^2

(2) $3\sqrt{3}$ m/s^2

(3) $6\sqrt{3}$ m/s^2

(4) $8\sqrt{3}$ m/s^2

14) Two trains one of length 100 m and another of length 125 m, are moving in mutually opposite directions along parallel lines. They meet each other when each train has speed of 10 m/s. If their acceleration are 0.3 m/s^2 and 0.2 m/s^2 respectively, then the time they take to pass each other will be :-

(1) 5 s

(2) 10 s

(3) 15 s

(4) 20 s

15) A body starts from rest with uniform acceleration. If its velocity after n second is v, then its displacement in the last two seconds is :-

(1) $\frac{2v(n+1)}{n}$

(2) $\frac{v(n+1)}{n}$

(3) $\frac{v(n-1)}{n}$

(4) $\frac{2v(n-1)}{n}$

16) When a car is at rest, its driver sees rain drops falling on it vertically. When driving the car with speed v, he sees that rain drops are coming at an angle 60° from the horizontal. On further increasing the speed of the car to (1 + β)v, this angle changes to 45°. The value of β is close to :

(1) 0.41

(2) 0.50

(3) 0.37

(4) 0.73

17) A man can swim in still water at a speed of 5 km/hr. He wants to cross a river 6 km wide, flowing at the rate of 4 km/hr. If he heads in a direction making an angle of 127° with stream direction, then he will reach a point on the other bank (approx) :-

(1) upstream at a distance of 1.5 km

(2) downstream at a distance of 1.5 km

(3) directly on the other side of the bank

(4) never reach the other bank

18) The velocity of a particle is given by $v = \sqrt{169-8x}$, where x is position in metre and v in m/s. The acceleration of the particle will be

(1) 7 m/s^2

(2) –4 m/s^2

(3) –8 m/s^2

(4) –2 m/s^2

19) A person hired a boat for 3 hrs. The river flows at a speed of 2.5 km/hr and speed of boat is 7.5 km/hr w.r.t. water. They start from a point rowing along the river and comes back to the same point in 3 hrs. Then distance travelled by him is :

(1) 20 km

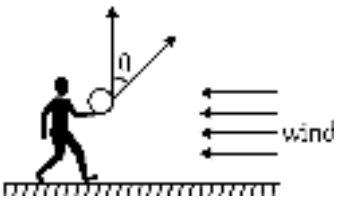
(2) 8 km

(3) 6 km

(4) 5 km

20)

A boy throws a ball upwards with velocity v_0 = 20 m/s. The wind imparts a horizontal acceleration of 4 m/s^2 to the ball. The angle θ from vertical at which the ball must be thrown so that the ball returns to the boy's hand is (g = 10 m/s^2)



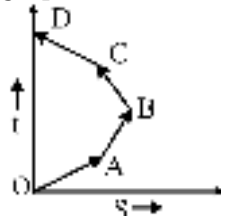
(1) $\tan^{-1}(1.2)$

(2) $\tan^{-1}(0.2)$

(3) $\tan^{-1}(2)$

(4) $\tan^{-1}(0.4)$

21) Which of the following options is correct for the object having a straight line motion represented by the following graph :-



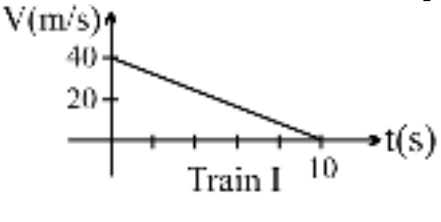
(1) The object moves with constantly increasing velocity from O to A and then it moves with constant velocity.

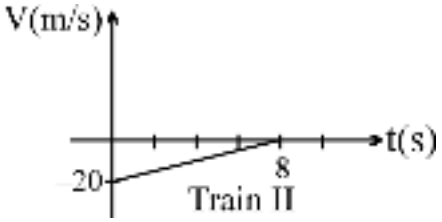
(2) Velocity of the object increases uniformly

(3) Average velocity is zero

(4) The graph shown is impossible

22) Two trains, which are moving along different tracks in opposite directions towards each other, are put on the same track by mistake. Their drivers, on noticing the mistake, start slowing down the trains when the trains are 300 m apart. Graphs given below show their velocities as function of time as the trains slow down. The separation between the trains after both have stopped, is :





(1) 120 m

(2) 280 m

(3) 60 m

(4) 20 m

23) Two particles were thrown simultaneously from the same point one straight up, and the other at an angle of $\theta = 30°$ to the horizontal. The initial velocity of each particle in equal to $V_0 = 25$ m/s neglecting air drag. Find the distance between the particles 1 sec after projection :-

(1) $\frac{25\sqrt{3}}{2}$ m

(2) 25m

(3) 12.5m

(4) 15m

24) A car starting from rest, accelerates at the rate f through a distance d, then continuous at constant speed for time t and then decelerates at the rate $\frac{f}{2}$ to come to rest. If the total distance traversed is 5d, then :-

(1) $d = \frac{1}{4}ft^2$

(2) $d = \frac{1}{2}ft^2$

(3) $d = \frac{1}{6}ft^2$

(4) d = ft

25) The equation of projectile is $y = x\sqrt{3} - \frac{x^2}{5}$ (origin is point of projection); (x → horizontal; y → vertical)

Column-I		Column-II	
A.	Initial velocity	**P.**	$5\sqrt{3}$ S.I. unit
B.	Range of projectile	**Q.**	10 S.I. unit
C.	Maximum height of projectile	**R.**	3.75 S.I. unit
D.	Angle of projection	**S.**	$\frac{\pi}{3}$ S.I. unit

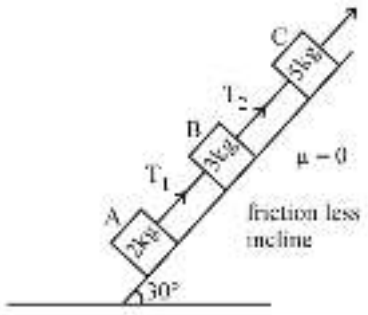
(1) A-Q; B-P; C-R; D-S

(2) A-R; B-S; C-P; D-Q

(3) A-P; B-R; C-S; D-Q

(4) A-S; B-P; C-Q; D-R

26) Three blocks A, B, C of mass 2kg, 3kg, 5kg are placed on a frictionless incline as shown in figure. Force F of magnitude 150 N is applied on block 'C' as shown. Find tension in string 1(T_1) :-



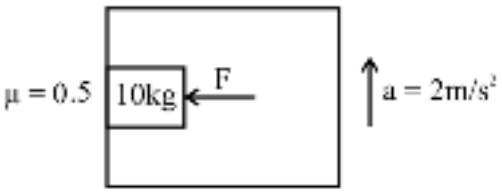
(1) 20 N

(2) 30 N

(3) 10 N

(4) 40 N

27) A block of mass 10 kg is hold on a vertical wall of lift which is going upward with acceleration of 2 m/s^2. Find minimum value of F so that we can hold the block w.r.t. lift. Friction coffecient between



block & wall of lift is 0.5 :-

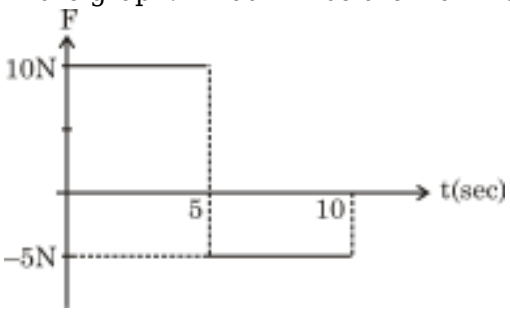
(1) 40 N

(2) 50 N

(3) 240 N

(4) 60 N

28) A particle of mass 2kg moving with speed 2m/s is acted up by force F, whose variation is shown in the graph. What will be the maximum speed of particle during the subsequent motion ?



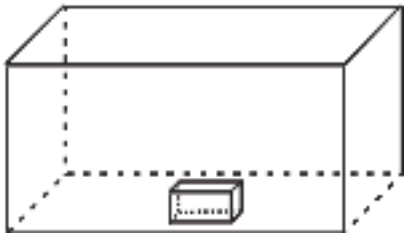
(1) 25 m/s

(2) 27 m/s

(3) 20 m/s

(4) 23 m/s

29) A solid block of mass 2 kg is resting inside a cube as shown in Fig. The cube is moving with a velocity $\vec{v} = 5\hat{i} + 2\hat{j}\,ms^{-1}$. If the coefficient of friction between the surface of cube and block is 0.2, then the force of friction between the block and cube is :-



(1) 10 N

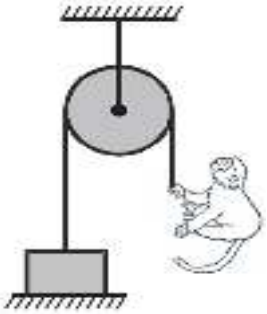
(2) 4 N

(3) 14 N

(4) Zero

30) A monkey weighing 10 kg is climbing up a light rope which passes over an ideal pulley. The

other end of the rope is attached a 15 kg mass as shown in the figure. In order to raise the 15 kg mass off the ground the monkey should climb up :



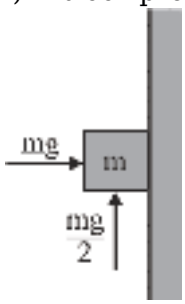
(1) with constant acceleration g/3

(2) with an acceleration greater than g/2

(3) with an acceleration greater than g/4

(4) It is not possible because weight of monkey is lesser than the block

31) A block pressed against the vertical wall is in equilibrium. The minimum coefficient of friction is :-



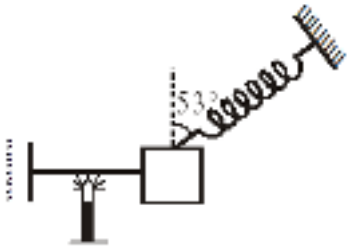
(1) 0.4

(2) 0.2

(3) 0.5

(4) None of these

32) The block shown in the figure is in equilibrium. Find acceleration of the block just after the string burns.



(1) $\frac{3g}{5}$ m/s^2

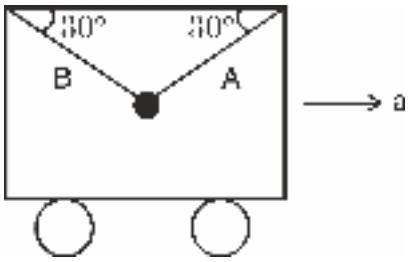
(2) $\frac{4g}{5}$ m/s^2

(3) $\frac{4g}{3}$ m/s^2

(4) None of these

33) A car is moving with uniform acceleration 'a' and a particle is suspended with the help of two

cords A and B as shown. If tension in the cord A is twice the weight of particle, then 'a' will be :



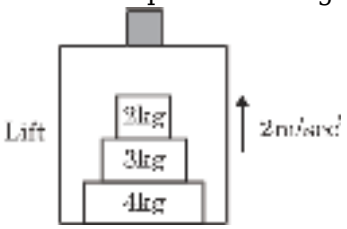
(1) $\sqrt{3}g\ m/s^2$

(2) $\frac{g}{2}\ m/s^2$

(3) $\frac{g}{3\sqrt{3}}\ m/s^2$

(4) $\frac{g}{2\sqrt{3}}\ m/s^2$

34) A lift is moving upwards with an acceleration of 2 m/sec^2. Inside the lift a 4 kg block is kept on the floor. On the top of it, 3 kg block is placed and again a 2 kg block is kept on the 3 kg block.



Calculate contact force between 2 kg block and the 3 kg block.

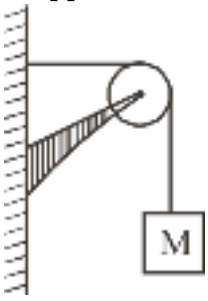
(1) 24 N

(2) 48 N

(3) 12 N

(4) 36 N

35) A string of negligible mass going over a clamped pulley of mass m supports a block of mass M



as shown in the figure. The force on the pulley by the clamp is given.

(1) $\sqrt{2}Mg$

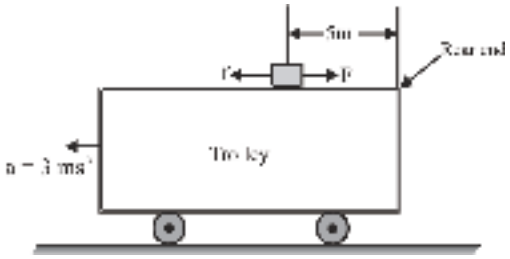
(2) $\sqrt{2}mg$

(3) $\sqrt{(M+m)^2+m^2}\,g$

(4) $\sqrt{(M+m)^2+M^2}\,g$

36) A block of mass 10 kg is placed at a distance of 5 m from the rear end of a long trolley as shown in fig. The coefficient of friction between the block and the surface below is 0.2. Starting from rest,

the trolley is given a uniform acceleration of 3 ms^{-2}. At what distance from the starting point will the block fall off the trolley- (Take g = 10 ms^{-2})



(1) 15 m

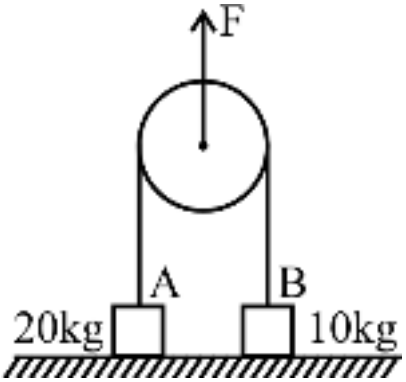
(2) 20 m

(3) 25 m

(4) 30 m

37)

In the figure shown blocks are initially at rest on the floor. Mark the correct statement :



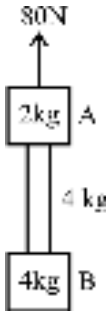
(1) When applied force on pulley is 200 N the normal reaction on block A is zero.

(2) When applied force is 400 N acceleration of B is 10 m/s^2

(3) When applied force is 240 N acceleration of pulley is 2 m/s^2

(4) For a force F = 100 N, normal reaction on B is zero.

38) Two objects A and B of masses 2 kg and 4kg are connected by a uniform rope of mass 4kg as shown in the diagram. A force of magnitude 80N acts on A in vertically upward direction. Tension at mid point of the rope is (g = 10 m/s^2) :-



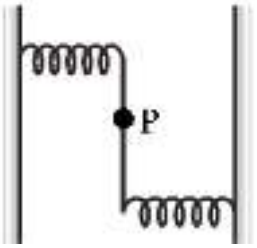
(1) 32 N

(2) 40 N

(3) 42 N

(4) 48 N

39) A uniform rod of length L and mass M is pivoted at the centre. Its two ends are attached to two springs of equal spring constants k. The springs are fixed to rigid supports as shown in the figure, and the rod is free to oscillate in the horizontal plane. The rod is gently pushed through a small angle θ in one direction and released. The frequency of oscillation is :-



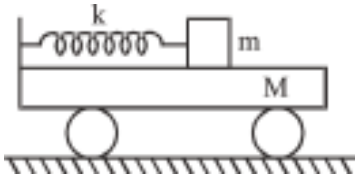
(1) $\frac{1}{2\pi}\sqrt{\frac{2k}{M}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{k}{M}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{6k}{M}}$

(4) $\frac{1}{2\pi}\sqrt{\frac{24k}{M}}$

40) Initially spring is stretched by x_0 and released. Find time period of oscillation :-



(1) $2\pi\sqrt{\frac{M}{k}}$

(2) $2\pi\sqrt{\frac{M+m}{k}}$

(3) $2\pi\sqrt{\frac{mM}{k(M+m)}}$

(4) $2\pi\sqrt{\frac{M-m}{k}}$

41) Vertical displacement of a plank with a body of mass 'm' on it is varying according to law $y = \sin \omega t + \sqrt{3} \cos \omega t$. The minimum value of ω for which the mass just breaks off the plank and the moment it occurs first after t = 0 are given by : (y is positive vertically upwards)

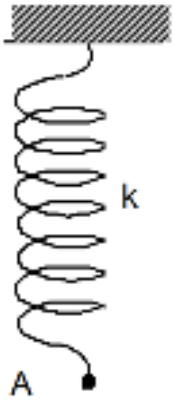
(1) $\sqrt{\frac{g}{2}}, \frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}}$

(2) $\frac{g}{\sqrt{2}}, \frac{2}{3}\sqrt{\frac{\pi}{g}}$

(3) $\sqrt{\frac{g}{2}}, \frac{\pi}{3}\sqrt{\frac{2}{g}}$

(4) $\sqrt{2g}, \sqrt{\frac{2\pi}{3g}}$

42) A light spring of stiffness k is hanging from the ceiling. It is stretched by a distance d, then a mass m attached to the end A which is then released. The amplitude of the oscillations will be



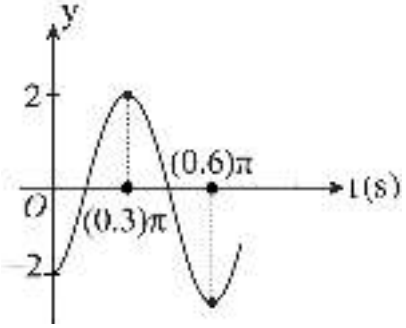
(1) mg/k

(2) 2mg/k

(3) d

(4) |(mg/k) - d|

43) Part of a simple harmonic motion is graphed in the figure, where y is the displacement from the mean position. The correct equation describing this S.H.M is :



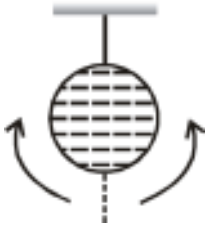
(1) y = 4 cos (0.6t)

(2) $y = 2 \sin\left(\frac{10}{3}t - \frac{\pi}{2}\right)$

(3) $y = 4 \sin\left(\frac{10}{3}t + \frac{\pi}{2}\right)$

(4) $y = 2 \cos\left(\frac{10}{3}t + \frac{\pi}{2}\right)$

44) A simple pendulum is made of a hollow sphere containing mercury suspended by means of a wire and has hole at bottom. If mercury leaks out through the hole, the period of pendulum will :-



(1) Remains unchanged

(2) First increase then decrease

(3) First decrease then increase

(4) Become erratic

45) A simple pendulum performs simple harmonic motion about X = 0 with an amplitude A and time period T. The half value of speed of the pendulum at X = $\frac{A}{2}$ will be :-

(1) $\frac{\pi A\sqrt{3}}{T}$

(2) $\frac{\pi A}{T}$

(3) $\frac{\pi A\sqrt{3}}{2T}$

(4) $\frac{3\pi^2 A}{T}$

CHEMISTRY

1) Which one of the following orders correctly represents the increasing acid strengths of the given acids :

(1) $HOClO_3 < HOClO_2 < HOClO < HOCl$

(2) $HOCl < HOClO < HOClO_2 < HOClO_3$

(3) $HOClO < HOCl < HOClO_3 < HOClO_2$

(4) $HOClO_2 < HOClO_3 < HOClO < HOCl$

2) Which of the following species are amphoteric :
(I) $Al(OH)_3$ (II) $Ba(OH)_2$ (III) $Zn(OH)_2$

(1) I only

(2) I, II only

(3) I & III only

(4) II & III only

3) Chalcogens are elements of :-

(1) Group 16

(2) p block

(3) $ns^2\ np^4$ configuration

(4) All are correct

4) Ionic potential shows :-

(1) Ionization potential

(2) Charge to radius ratio of a cation

(3) Quantum of charge on cation

(4) Charge to radius ratio of anion

5) Which of the following is **CORRECT** order :-

(1) $\ddot{N}H_3 > \ddot{P}H_3 > \ddot{A}sH_3 > \ddot{S}bH_3$ → Basic strength

(2) HF < HCl < HBr < HI → Acidic strength

(3) NaOH < KOH < RbOH < CsOH→ Basic strength

(4) All of these

6) Match the column

Column-I		**Column-II**	
(Atomic number)		**(Position of element in Periodic table)**	
(A)	Z = 37	(P)	p-block
(B)	Z = 42	(Q)	f-block
(C)	Z = 34	(R)	d-block
(D)	Z = 92	(S)	s-block

(1) A-P, B-Q, C-S, D-R

(2) A-P, B-Q, C-R, D-S

(3) A-S, B-R, C-P, D-Q

(4) A-S, B-R, C-Q, D-P

7) Which of the following electronic configuration does not represent any block element :-

(1) $(n-1)\ d^{1-10}\ ns^{1-2}$ (n = 4, 5, 6...)

(2) $(n-2)\ f^{14}\ (n-1)\ d^{2-10}\ ns^{1,2}$ (n = 6)

(3) $(n-1)\ d^{1-10}\ ns^{0,1,2}$ (n = 5....)

(4) $(n-1)\ d^{1-10}\ ns^{1-2}$ (n = 3)

8) Which of the following is correct :-

(1) Sodium is a p-block element

(2) p-orbital contains 6 electrons

(3) Z_{eff} increases in a period (left of right)

(4) Both 2, 3

9) Electronic configuration of X^{2+} and Y^{3+} are : X^{2+} = [Ar]$3d^8$,Y^{3+}=[Ar]$3d^3$. What are the atomic number of X and Y respectively ?

(1) 28, 24
(2) 28, 25
(3) 28, 26
(4) 28, 27

10) Which pair of element is different from others :-

(1) Li, Na
(2) B,Si
(3) Li,Mg
(4) Be,Al

11)

Match of the formulas given in Column-I with the type of oxide given in column-II and mark the correct option :-

Column-I		Column-II	
(A)	Pb_3O_4	(I)	Neutral oxide
(B)	N_2O	(II)	Acidic oxide
(C)	Mn_2O_7	(III)	Basic oxide
(D)	Bi_2O_3	(IV)	Mixed oxide

(1) A-I, B-II, C-III, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-III, C-I, D-II

12)

Xe has maximum electronegativity in

(1) XeF_2
(2) XeF_4
(3) XeF_6
(4) None of the above

13) Which of the following E.A. order is not correct ?

(1) N < O < S < F

(2) Cl > O > N > C

(3) O < S < F < Cl

(4) B < C < Si < S

14) If ionic radius of Al^{+3} = 0.5 Å, Mg^{+2} = 0.72 Å & K^{+} = 1.33 Å then Li^{+} = ?

(1) 1 Å

(2) 0.5 Å

(3) 0.69 Å

(4) 0.74 Å

15) Correct order of I.P (1st) is ?

(1) F < S < P < Na

(2) P < S < Na < F

(3) Na < P < S < F

(4) Na < S < P < F

16) IP of an element is x and EA is y then correct relation is :-

(1) x < y

(2) x = y

(3) x > y

(4) None

17) Exothermic process is ?

(1) $S^{\ominus}_{(g)} \rightarrow S^{-2}_{(g)}$

(2) $N_{(g)} \rightarrow N^{\ominus}_{(g)}$

(3) $Al^{+2}_{(g)} \rightarrow Al^{+3}_{(g)}$

(4) $Na^{+}_{(g)} + Cl^{\ominus}_{(g)} \rightarrow NaCl_{(s)}$

18) Which of the following represents the first I.P. of given element ?

(1) $Na_{(s)} \rightarrow Na^{+}_{(g)}$

(2) $F_{2(g)} \rightarrow 2F^{+}_{(g)}$

(3) $Mg_{(g)} \rightarrow Mg^{+2}_{(g)}$

(4) $Al_{(g)} \rightarrow Al^{+}_{(g)}$

19) Correct order of 2nd IP :-

(1) Cu < Zn

(2) K > Cu

(3) K < Cu

(4) N > O

20) Which of the following oxide does not react with NaOH as well as with HCl :-

(1) Al_2O_3

(2) ZnO

(3) N_2O

(4) All

21) Consider the reaction : $2N_2O_4 \rightleftharpoons 4NO_2$.

If $\frac{-d[N_2O_4]}{dt} = k$ and $\frac{d[NO_2]}{dt} = k'$ then

(1) 2k' = k

(2) k' = 2k

(3) k' = k

(4) $k = \frac{1}{4}k'$

22) What would be the rate law of the reaction

A + B → Products

If following data is collected experimentally ?

Exp.	Initial conc. [A]	Initial conc. [B]	Initial rate (r)
1	0.1 M	1.0 M	2.1×10^{-3}
2	0.2 M	1.0 M	8.4×10^{-3}
3	0.2 M	2.0 M	8.4×10^{-3}

(1) $r = k[A]^2$

(2) $r = k[B]^2$

(3) $r = k[A]^2 [B]^1$

(4) $r = k[A]^1 [B]^1$

23) Mechanism of reaction $2O_{3(g)} \rightarrow 3O_{2(g)}$ can be given as

$O_3 \underset{K_2}{\overset{K_1}{\rightleftharpoons}} O_2 + [O]$ (fast)

$O_3 - O \xrightarrow{K_3} 2O_2$ (slow)

Which of the following is correct rate law :-

(1) $r = K_3[O_3] [O]$

(2) $r = \frac{K_1K_3}{K_2} [O_3]^2 [O_2]^{-1}$

(3) $r = K_1K_2 [O_3] [O_2]$

(4) $r = \frac{K_3K_1}{K_2}[O_2]^2[O_3]^{-1}$

24) For an elementary reaction which of the following is not true :-

(1) Order is not equal to the molecularity

(2) Elementary reactions take place in one step

(3) Elementary reactions can not have molecularity > 3

(4) Elementary reactions are also called simple reactions

25) Graph plotted between $[A]_t$ vs t for zero order reaction is straight line with intercept 0.003 and slope –0.48. then what will be value of rate constant (Ms^{-1}) ?

(1) 3×10^{-3}

(2) 0.48

(3) 0.24

(4) 0.72

26) 80% of a first order reaction completes in 70 min. How much time it will take to complete 90% of reaction ?

(1) 114 min.

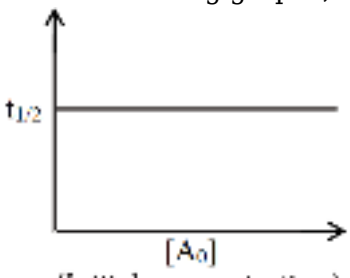
(2) 140 min.

(3) 100 min.

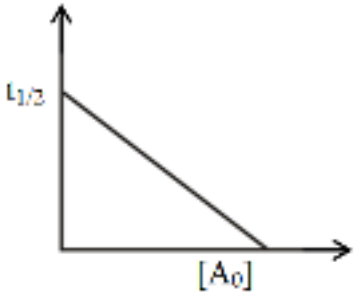
(4) 200 min.

27) In the following graphs, which graph(s) belong to first order reaction :

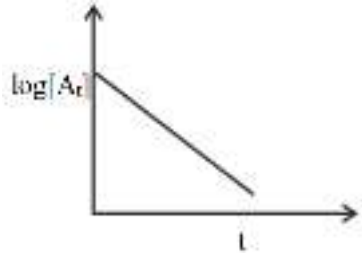
(i)



(ii)



(iii)



(1) (i), (iii)

(2) (i), (ii), (iii)

(3) (i) only

(4) (iii), (ii)

28)

For a first order homogeneous gaseous reaction A → 3B, if pressure after time t was P_T and initial pressure was P^0 then select correct relation :-

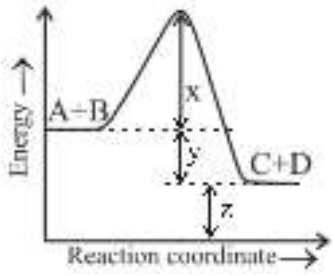
(1) $k = \frac{1}{t} \ln \left[\frac{P^0}{3(P^0 - P_T)} \right]$

(2) $k = \frac{1}{t} \ln \left[\frac{2P^0}{3(P^0 - P_T)} \right]$

(3) $k = \frac{1}{t} \ln \left(\frac{3P^0}{2P^0 - P_T} \right)$

(4) $k = \frac{1}{t} \ln \left(\frac{2P^0}{3P^0 - P_T} \right)$

29) Following diagram is given for the reaction A + B → C + D. The enthalpy change and activation energy for the reverse reaction C + D → A + B are respectively



(1) x, y

(2) x, x + y

(3) y, x + y

(4) y, y + z

30) For a first order reaction, A → P, the temperature (T) dependent Rate constant (k) was found to follow the equation $\log k = 6 - \frac{1000}{2T}$. The value of activation energy 'E_a' will be :-

(1) 1 J/mole

(2) 9.56 kJ/mole

(3) 1 cal/mole

(4) 4.15 kcal/mole

31) **Assertion :** The catalyst does not affect the heat of reaction.
Reason : Catalyst increases the rate of reaction.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of assertion.

(3) If Assertion is true but reason is false.

(4) If both assertion and reason are false.

32) Which statements are true (T) and false (F) with reference to reaction $2A + B \rightarrow C$, rate = k[A][B]? (1) Value of k is independent of concentration of A and B.
(2) Unit of k is $time^{-1}$.
(3) Half life time is constant.
(4) Rate of formation of C is double than the rate of consumption of A.

(1) TFFF

(2) FFFT

(3) TTFF

(4) FFTT

33) The vapour pressure of water at 300 K in a closed container is 0.4 atm. If amount of water is doubled, its vapour pressure at the same temperature will be :

(1) 0.8 atm

(2) 0.2 atm

(3) 0.4 atm

(4) 0.6 atm

34) At what mole fraction of benzene the mixture of benzene and toluene will boil at 88ºC.
(At 88°C $P^{º}_{benzene}$ = 960 mm and $P^{°}_{toluene}$ = 380 mm)

(1) 0.345

(2) 0.25

(3) 0.5

(4) 0.655

35) Pure water boils at 373 K and nitric acid at 359 K. The mixture of water and nitric acid boils at 393.5 K. The mixture shows :

(1) No deviation from Raoult's law

(2) Positive deviation from Raoult's law

(3) Negative deviation from Raoult's law

(4) The solution is unsaturated

36) Addition of 4 g of a non-volatile solute to 54 g water results in a lowering of vapour pressure of water by 1%. What is the molar mass of non-volatile solute ?

(1) 133.33

(2) 66

(3) 132

(4) 333

37) At equilibrium mole fraction of benzene in vapour phase is 0.75 then vapour pressure of the ideal solution of benzene and toluene is :

(Given $P°_{benzene}$ = 75 mm of Hg, $P°_{toluene}$ = 25 mm of Hg)

(1) 625 mm

(2) 50 mm

(3) 30 mm

(4) 40 mm

38) In a solution the partial vapour pressure of methanol and ethanol are 2.8 and 4.2 kPa respectively, then find the mole fraction of methanol in the vapour at equilibrium is :

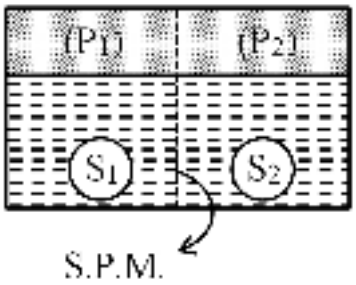
(1) 0.4

(2) 0.67

(3) 0.6

(4) 0.33

39) If vapour pressure (P_1) is lower than vapour pressure (P_2) in the given diagram. Then



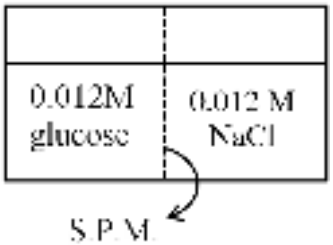
(1) More solvent will flow from S_1 to S_2

(2) More solvent will flow from S_2 to S_1

(3) Solvent from S_1 to S_2 will flow at equal rates

(4) No flow of solvent will take place

40) On which of these solutions must pressure be applied to prevent a net flow of water through the SPM in the given diagram.



(1) On the 0.012 M glucose solution

(2) On the 0.012 M NaCl solution

(3) Equal pressure on both solutions

(4) No pressure is required

41) The latent heat of vaporization of a liquid of molar mass 90 g/mol and B.P. 127°C is 9 Kcal/mol. The ebuillioscopic constant of liquids is (K-kg/mol)

(1) 3.2

(2) 0.04

(3) 0.32

(4) 0.52

42) Molal depression constant for Pb is 8.5°C kg/mol. How many g of Sn must be dissolved in 425 g of Pd to produce an alloy with melting point of 322°C ? The melting point of pure Pb is 327°C (At wt. Sn = 120)

(1) 30 g

(2) 15 g

(3) 86 g

(4) 41 g

43) **Assertion (A) :** Except osmotic pressure, all other colligative properties depend on the nature of solvent.
Reason (R) : Colligative properties are intensive properties.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

44)

	Column-I (Solute in water)		**Column-II (D.O.D. to make van't Hoff factor equal to 2)**
(A)	$K_4[Fe(CN)_6]$	(P)	0.33
(B)	KCl	(Q)	1
(C)	Na_2SO_4	(R)	0.5
(D)	$K_3[Fe(CN)_6]$	(S)	0.25

	A	B	C	D
(1)	P	Q	R	S
(2)	S	R	Q	P
(3)	S	Q	R	P
(4)	S	Q	P	R

(1) 1

(2) 2

(3) 3

(4) 4

45)

'a' moles of KCl and 'b' mole of $BaCl_2$ are dissolved in 1 kg of water. If (a + b) = 0.1, then the observed range of ΔT_f is :
(K_f = 1.86 k-kg mol^{-1} for water)

(1) 0.372°C to 0.93°C
(2) 0.93°C to 0.186°C
(3) 0.558°C to 0.372°C
(4) 0.372°C to 0.558°C

BIOLOGY

1) Which of the following set of organism have motile male gametes?

(1) *Fucus*, *Porphyra*, *Ulothrix*
(2) *Fucus*, *Chlamydomonas*, *Spirogyra*
(3) *Fucus*, *Gracillaria*, *Chlamydomonas*
(4) *Fucus*, *Chlamydomonas*, *Funaria*

2) Oogamous type of sexual reproduction reported in :-

(1) *Volvox*
(2) *Fucus*
(3) *Spirogyra*
(4) More than one option are correct

3) Which plant group has integumented megasporangium.

(1) Bryophyta
(2) Pteridophyta
(3) Thallophyta
(4) Spermatophyta

4) Choose the incorrect match from the following :-

(1) Cyanophyceae - Chlorophyll 'a', 'e', Starch & oil droplets
(2) Phaeophyceae - Chlorophyll 'a', 'c', Mannitol
(3) Chlorophyceae - Chlorophyll 'a', 'b', Starch
(4) Rhodophyceae - Chlorophyll 'a', 'd', Floridean starch

5) The free-living gametophytes of liverworts are formed due to germination of :-

(1) Gametes
(2) Spores
(3) Lateral buds
(4) Protonema

6) The first stage in the life cycle of moss, which develops directly from a spore is :-

(1) Prothallus
(2) Leafy stage
(3) Strobilus
(4) Protonema

7) **Assertion :** The members of rhodophyceae are commonly called red algae.
Reason : Red algae have predominance of the red pigment, r-phycoerythrin in their body.

(1) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion
(2) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion
(3) Assertion is correct but reason is incorrect
(4) Assertion is incorrect but Reason is correct

8) **Assertion :** In bryophytes, zygote produce a multicellular body called a sporophyte.
Reason : Zygote do not undergo reduction division immediately.

(1) Both Assertion & Reason are correct & Reason is correct explanation of the Assertion.
(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion.
(3) Assertion is correct but Reason is incorrect.
(4) Assertion is incorrect but Reason is correct.

9)

Which of the following is the correct sequence of structure formed during life cycle of Moss?

(1) Gametophyte → Buds → Protonema → Spores → Sporophyte
(2) Sporophyte → Spores → Protonema → Buds → Gametophyte
(3) Sporophyte → Zygote → Buds → Spores → Gametophyte
(4) Gametophyte → Zygote → Sporophyte → Embryo → Buds

10) Which of the following is not true for protonema of moss ?
(a) Creeping (b) Non-Green
(c) Unbranched (d) Filamentous
(e) Develops from gamete
Options :

(1) a and d only
(2) a, d and e only
(3) b and d only
(4) b, c and e only

11) A botany student encountered a palm like, short tree with pinnately compound leaves with sessile leaflets bearing mid rib but no lateral veins. Unbranched stem has persistent woody leaf bases. The plant has to be :-

(1) Cycas

(2) Pinus

(3) Dryopteris

(4) Cedrus

12) Which has vascular tissue and produce spores but does not has seeds ?

(1) Bryophyta

(2) Pteridophyta

(3) Gymnosperm

(4) Angiosperm

13) Consider the following four statements (a-d) and select the option which includes all the correct ones only:-
(a) In gymnosperms, the male and female cones are formed.
(b) In Bryophyte and pteridophyte sporophyte have an independent free living existence.
(c) *Chlorella* and *Spirulina* are rich in proteins and are used as food supplements.
(d) In Algae, bryophyte and pteridophytes, gametophyte have an independent free living existence.

(1) Statement b, c and d

(2) Statement a and b

(3) Statement c and d

(4) Statement a, c and d

14) In which plants male gametes are motile :-
(A) Bryophyta (B) Pteridophyta
(C) Angiosperm (D) Algae

(1) A & B

(2) B & C

(3) A, B & C

(4) A, B & D

15) A plant shows sporophyte as a main generation. Its gametophyte shows rhizoids and is haploid. It needs water to complete its life cycle because the male gametes are motile. Identify the group of to which it belons to

(1) Pteridophytes

(2) Gymnosperms

(3) Monocots

(4) Bryophytes

16) In conifers, transpiration is reduced due to the presence of :-
(I) Needle-like leaves
(II) Thin cuticle on roots
(III) Sunken stomata in leaves

(1) I only
(2) I and II
(3) II and III
(4) I and III

17) Motile stages are not found in life cycle of :-

(1) Red Algae & green Algae
(2) Red Algae & brown Algae
(3) Red Algae & blue green Algae
(4) Green Algae & brown Algae

18)

Which of the following statement about Cycas is correct :-
(a) Its coralloid roots contain some blue green algae
(b) It does not have female cone
(c) It has circinate Archigonia

(d) Its xylem is mainly composed of xylem vessels
(1) a, b and c are correct
(2) a and d are correct
(3) b and d are correct
(4) a, b and d are correct

19) More developed and elaborate sporophyte is found in -

(1) *Marchantia*
(2) *Sphagnum*
(3) *Riccia*
(4) *Ulothrix*

20) Agar-agar is obtained from :-

(1) *Gracilaria*
(2) *Gonyaulax*
(3) *Gelidium*
(4) Both (1) and (3)

21) Which is not found in bacteria ?

(1) Mitochondria
(2) DNA
(3) Ribosome
(4) Cytoplasm

22) According to the size of the organism, which sequence is not correct :

(1) Eukaryotic cell > Prokaryotic cell > PPLO > Virus

(2) Virus < PPLO < Prokaryotic cell < Eukaryotic cell

(3) Eukaryotic cell > PPLO > Prokaryotic cell > Virus

(4) PPLO < Prokaryotic cell < Eukaryotic cell < Tissue

23) Cell recognition and adhesion are facilitated by components of plasma membrane. These components are generally.

(a) Glycolipids (b) Glycoproteins

(c) Water (d) Cholesterol

(1) a and d

(2) a and c

(3) b and d

(4) a and b

24) Which of the following is true for nucleolus ?

(1) It is a double membrane bound structure.

(2) It takes part in spindle formation.

(3) It is a site for active ribosomal RNA synthesis

(4) It involve in oxidation of fats.

25) Find out the correct option regarding T. S. (Transvers Section) of cilia given below:

	Peripheral doublet microtubule	**Central microtubule**	**Radial spoke**	**Central sheath**
(1)	9	0	9	1
(2)	9	1	9	9
(3)	9	2	9	1
(4)	9	0	1	9

(1) 1

(2) 2

(3) 3

(4) 4

26) Find out the incorrect match :-

(1) Lysosome – Intracellular scavanging

(2) Golgi complex – Metabolism of xenobiotics

(3) Elaioplast – Oil and fat storage

(4) Glyoxysomes – Gluconeogenesis

27) Which of the following is not a part of cytoskeleton ?

(1) Microtubule

(2) Microfilament

(3) Microbodies

(4) Intermediate filament

28) Cellular organelles with membranes bound are :-

(1) Lysosomes, Golgi apparatus & mitochondria

(2) Nucleus, Ribosomes & mitochondria

(3) Chromosomes, Ribosomes & endoplasmic reticulum

(4) Endoplasmic reticulum, Ribosomes & Nuclei

29) Which of the following are similar in mitochondria, chloroplast & bacteria
(A) 80S Ribosome (B) Circular ds DNA
(C) Binary fission (D) Porin protein
(E) Stroma

(1) A, B, C & D only

(2) A, B, C, D & E

(3) B, C & D only

(4) A, B & C only

30)

Endoplasmic reticulum, Golgibody, Lysosome, Mitochondria, Nucleus, Ribosome, Chloroplast, Leucoplast, Chromoplast, Vacuole
How many above given cell organelle is single membrane bound?

(1) 4

(2) 3

(3) 5

(4) 2

31) Microtubules and microfilaments are proteinaceons structures form cytoskeleton filaments of the cell. Which one is not a function of cytoskelton :

(1) Mechanical support

(2) Motility

(3) Maintenance of the shape of the cell

(4) Protein synthesis

32) Which of the following statements is not correct?

(1) The fluid nature of membrane is important for cell growth and cell division

(2) Plasma membrane is mainly composed of lipids + proteins

(3) Structure of plasma membrane was studied only after the advent of the electron microscope

(4) Integral proteins either lie on the surface or totally buried in the membrane

33) **Assertion :** In eukaryotic cells there is an extensive compartmentalisation of cytoplasm.
Reason : Eukaryotic cells possess membrane bound organelles.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

34) Match the column-I with column-II and choose right option regarding them.

Column-I		**Column-II**	
A	Photorespiration	i	Chromosome
B	Axoneme	ii	Centriole
C	Cart wheel structure	iii	Peroxisome
D	Centromere	iv	Cilia

	A	B	C	D
(1)	iv	iii	ii	i
(2)	iii	iv	i	ii
(3)	i	iii	iv	ii
(4)	iii	iv	ii	i

(1) 1

(2) 2

(3) 3

(4) 4

35) Read the following four statements (A - D) :-
(A) The interphase lasts more than 95% of the duration of cell cycle.
(B) In animal cells during the s-phase, DNA replication begins in the nucleus and the centriole duplicates in the nucleolus
(C) Since the number of chromosomes in the parent and progeny cells is the same so mitosis is also called as equational division.
(D) Mitosis usually results in the production of diploid daughter cells with identical genetic complement.
How many of the above statement/s is/are incorrect?

(1) One

(2) Two

(3) Four

(4) Three

36) Some events of mitotic cell division are given below in unordered manner -
(i) Condensation of chromatin is completed

(ii) Chromosome lose their identity as discrete elements
(iii) Sister chromatid separate
(iv) Initiation of the assembly of mitotic spindle between pair of centrioles
The correct order for steps of cell division is :-

(1) i → iv → iii → ii
(2) iv → i → iii → ii
(3) ii → i → iv → iii
(4) iv → iii → i → ii

37) In which stage of cell division, chromosomal structure is best observed ?

(1) Prophase
(2) Interphase
(3) Metaphase
(4) Telophase

38) If mitotically dividing vegetative cell of flowering plant at G_1 stage has 20 chromosome with 20pg DNA then in anaphase cell will has :-

(1) 20 chromosomes/40pg DNA
(2) 40 chromosomes/40pg DNA
(3) 40 chromosomes/80pg DNA
(4) 10 chromosomes/40pg DNA

39) In which phase of meiosis sister chromatid are separate?

(1) Anaphase - I
(2) Metaphase - I
(3) Diplotene
(4) Anaphase - II

40) The beginning of diplotene is recognised by :-

(1) Terminalisation of chiasmata
(2) Disappearence of nucleolus and nuclear membrane
(3) Dissolution of synaptonemal complex
(4) Recombination of genetic material on the two chromosome

41) Duplication of centriole takes place in :-

(1) G_1 phase
(2) S-phase
(3) G_2 phase
(4) M-phase

42) Which is the correct order of cell cycle ?

(1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow$ Prophase $\rightarrow$ Metaphase $\rightarrow$ Anaphase $\rightarrow$ Telophase $\rightarrow$ cytokinesis

(2) $G_2 \rightarrow G_1 \rightarrow S \rightarrow P–M \rightarrow A \rightarrow T \rightarrow$ Cytokinesis

(3) $S \rightarrow G_1 \rightarrow G_2 \rightarrow P–M \rightarrow A \rightarrow T \rightarrow$ Cytokinesis

(4) Prophase $\rightarrow$ Metaphase $\rightarrow$ Anaphase $\rightarrow$ Telophase $\rightarrow$ Cytokinesis $\rightarrow S \rightarrow G_2 \rightarrow G_1$

43) Given below are two statement :- **Statement-I :** In animal cells, the centrosome undergoes duplication during S-phase of interphase.
Statement-II : After S-phase, the number of chromosomes becomes double but amount of DNA remains the same. In the light of above statements, choose the most appropriate answer from the options given below :-

(1) Both statement I and statement II are incorrect

(2) Statement I is correct but and statement II is incorrect

(3) Statement I is incorrect but and statement II is correct

(4) Both statements I and statement II are correct

44) A loose and indistinct network of nucleoprotein fibres can be observed in :-

(1) Late prophase

(2) Metaphase

(3) Anaphase

(4) Interphase nucleus

45) Colchicine, an alkaloid obtained from a plant (Colchicum autumnale) is used to induce polyplody in plants. It increases the ploidy level by arresting formation of spindle fibres. At which stage, it affects the mitotic division?

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

46) Select the correctly written scientific name of Mango which was first described by Linnaeus-

(1) *Mangifera indica*

(2) *Mangifera indica* Linn.

(3) *mangifera indica*

(4) *Mangifera Indica.*

47) In the names *Mangifera indica, Solanum tuberosum* and *Panthera leo*, the words *indica*, *tuberosum* and *leo* represent :-

(1) Generic name

(2) Generic epithet

(3) Name of species

(4) Specific epithet

48) Which of the following are non-vascular cryptogames ?

(1) Algae and fungi

(2) Algae & Bryophyta

(3) Algae, Bryophyta & pteridophyta

(4) Only pteridophyta

49) A collection of species, which bears a close resemblance to one another in the morphological characters of the floral parts is known as :-

(1) Family

(2) Genus

(3) Division

(4) Variety

50) The relation of Solanaceae and Convolvulaceae with Polymoniales is similar to the relation occuring in :-

(1) Felidae and Canidae with Carnivora

(2) Primata and Carnivora with Mammalia

(3) Amphibia and Reptilia with Chordata

(4) *Solanum* and *Petunia* with Solanaceae

51) Classification of plants based on evolutionary relationship is called :

(1) Artificial system

(2) Natural system

(3) Phylogenetic system

(4) Numerical taxonomy

52) The classification system of Linnaeus was mainly based on the androecium structure therefore it was a/an :-

(1) Natural classification

(2) Numerical classification

(3) Artificial classification

(4) Phylogenetic classification

53) Arrange the following taxonomic groups (a,b,c and d) according to their taxonomic ranks in descending manner :-
(a) *Triticum* (b) Monocotyledonae
(c) Poaceae (d) Poales

(1) b → d → c → a

(2) b → c → a → d
(3) a → b → c → d
(4) d → c → b → a

54) Sole members of kingdom monera are-

(1) Diatoms
(2) Desmids
(3) Bacteria
(4) Lichens

55) Choose the correct sequence of taxonomial categories :-

(1) Species → Genus → Kingdom → Division
(2) Genus → Species → Kingdom → Division
(3) Species → Genus → Family → Order
(4) Species → Family → Genus → Order

56) Two kingdom system of classification is mainly based on:-

(1) Cell wall
(2) Cell membrane
(3) Mode of nutrition
(4) Phylogeny

57) Which option is not related to *mycoplasma* ?

(1) Penicillin resistant bacteria
(2) Faculative anaerobic, can survive without oxygen
(3) Lack cell membrane
(4) Pathogenic to plants and animals

58) Which of the following help in DNA replication in bacterial cell ?

(1) Lipopolysaccharide wall
(2) Peptidoglycan wall
(3) Ribosomes
(4) Mesosome

59) Heterocysts are related with :-

(1) Photosynthesis
(2) N_2 fixation
(3) Sexual reproduction
(4) All of these

60) Choose the incorrect statement from following:-

(1) Yeast are used to make bread and bear.

(2) Fungi prefer to grow in warm and humid places.

(3) Cell wall of fungi is composed of chitin and polysaccharides.

(4) Fungi show symbiotic association with algae as mycorrhiza and with roots of higher plants as lichens.

61) Read the following point :-
(A) Body septate and branched
(B) Some members are saprophytes or parasites
(C) Help in mineral cycling
(D) Reproduce by asexual spores known as conidia
Above points are related with which class of organisms

(1) Phycomycetes

(2) Euglenoids

(3) Deuteromycetes

(4) Basidiomycetes

62) Fruiting body forms during sexual reproduction of ?

(1) Ascomycetes

(2) Basidiomycetes

(3) Deuteromycetes

(4) Both (1) and (2)

63)

Find the **incorrect** match :-

Column-I		**Column-II**
(1)	*Ustilago*	belongs to class Basidiomycetes
(2)	*Agaricus*	belongs to class Basidiomycetes
(3)	*Trichoderma*	sexual cycle is found
(4)	*Alternaria*	sexual cycle is not found

(1) 1

(2) 2

(3) 3

(4) 4

64) Which of the following statement is not true :-

(1) Basidiomycetes ⇒ Bracket fungi are epixylic fungi i.e. these like to grow on wood

(2) Basidiomycetes ⇒ Puff balls are saprophytic Fungi. Fruiting body of puff ball bursts violently to release black ascospores

(3) Ascomycetes ⇒ *Neurospora* is used for the study of genetics in plant kingdom

(4) Ascomycetes ⇒ *Aspergillus flavus*-It prefers to grow on stored food and secretes toxic substances which are known as aflatoxins

65) Choose the odd ones from the followings with respect to absence of dikaryophase during sexual reproduction.
(a) Basidiomycetes (b) Ascomycetes
(c) Oomycetes (d) Phycomycetes

(1) a, b

(2) b, c

(3) c, d

(4) a, d

66) Read the following statements and find out which one is incorrect :-

(1) Virus take over the machinery of the host cell to replicate themself

(2) Virus contain both RNA and DNA

(3) AIDS in human is caused by a virus

(4) Generally in plant viruses, ss RNA is present

67) Mycorrhiza is :-

(1) A fungus parasiting root system of higher plants.

(2) An association of *rhizobium* with roots of leguminous plants

(3) A symbiotic association of plant roots and certain fungi

(4) An association of algae with fungi

68) Read the statements (A-D)
(A) Golden algae are also known as desmid.
(B) Mesosome is the invagination of cell membrane, responsible for oxidative phosphorylation, secretion and cell wall formation in bacteria.
(C) Archaebacteria differ from bacteria and eukaryotic plant cell in composition of cell wall
(D) Green pigments containing membranous extensions in cyanobacteria are heterocysts
How many of the above statements are correct ?

(1) Four

(2) Three

(3) Two

(4) One

69) Match Column-I with Column-II and select the correct option from the codes given the :-

Column-I	**Column-II**

A.	Chief producers in oceans	i.	Euglenoids
B.	Red tides	ii.	Diatoms
C.	Mixotrophic nutrition	iii.	Slime moulds
D.	Plasmodium	iv.	Dinoflagellates

(1) A-(ii),B-(iv), C-(i), D-(iii)

(2) A-(ii),B-(iv), C-(iii), D-(i)

(3) A-(ii),B-(iii), C-(i), D-(iv)

(4) A-(i),B-(iv), C-(iii), D-(ii)

70) Find out the mismatch :-

(1) Plasmogamy - Fusion of protoplasm of two gametes

(2) Karyogamy - Fusion of two nuclei

(3) Dikaryon - Two nuclei per cell

(4) Dikaryo phase - 2n condition

71) In flowering plants, the generative cell of the pollen grain

(1) Is bigger with abundant food reserve

(2) Has a large irregularly shaped nucleus

(3) Undergoes meiotic division to form gametes

(4) Is spindle shaped with dense cytoplasm and a nucleus

72) Which of following statement is/are correct ?

(1) The inner most layer of microsporangium is tapetum & it nourishes the developing pollen grains.

(2) Cells of tapetum generally have more than one nucleus.

(3) Exine of pollen grain has prominent apertures called germ pores.

(4) All the above

73) Perisperm is found in -

(1) Pea

(2) Mango

(3) Castor

(4) Gram

74) Which one is correct option regarding apocarpous gynoecium?

(1) Rose, *Michelia*, *Papaver*

(2) *Papaver*, *Hibiscus*, Lotus

(3) Rose, Lotus, *Michelia*

(4) *Papaver, Michelia*, Lotus

75) In over 60% of angiosperms, pollengrains are shed at ..... stage whereas in remaining angiosperms pollen grains are shed at 3-celled stage.

(1) 3-celled
(2) 2-celled
(3) 5-celled
(4) 1-celled

76) In a fertilized ovul n,2n and 3n conditions occurs respectively in -

(1) Antipodal, zygote and endosperm
(2) egg, nucellus and endosperm
(3) endosperm and egg.
(4) antipodal, synergids and integuments

77) The functional megaspore undergoes-

(1) 2 meiotic division to form mature embryo sac
(2) 3 mitotic division to form mature embryo sac
(3) 2 mitotic division to form mature embryo sac
(4) 3 meiotic division to form mature embryo sac

78) If diploid chromosomes number in a flowering plant is 12, then which one of the following will have only 6 chromosomes?

(1) Endosperm
(2) Leaf cells
(3) Cotyledons
(4) Synergids

79) Male and female flowers are present on same plant in all of these, except

(1) Cucurbits
(2) Coconuts
(3) Maize
(4) Date palm

80) Polyembryony is found in

(1) Citrus
(2) Mango
(3) *Ricinus*
(4) 1 and 2

81) A flower is large, colourful, fragrant and rich in nector. That flower is pollinated by :

(1) Snake
(2) Water
(3) Wind
(4) Insect

82) Which statement is incorrect

(1) Embryo develops at the micropylar end of embryo sac.
(2) Polar Nuclei are situated below the egg apparatus
(3) in Apomixis seeds are produced by fertilization
(4) Intine is made up of cellulose & Pectin

83) **Assertion :** Flowering plants have developed many out breeding devices to discourage self pollinations and to encourage cross-pollination.
Reason : Continued self-pollination result in inbreeding depression.

(1) Assertion is true but reason is false
(2) Both assertion & reason are true but reason is not the correct explanation of the assertion.
(3) Both assertion & reason are false.
(4) Both assertion & reason are true & the reason is the correct explanation of the assertion.

84)

Match the column-I with column-II.

	Column-I		**Column-II**
(A)	Sporopollenin	(i)	Synergids
(B)	Special cellular thickening	(ii)	Cellular endosperm
(C)	Nectar	(iii)	Exine of pollen grains
(D)	White kernel of coconut	(iv)	Floral reward

(1) A-iii ; B-ii ; C-iv ; D-i
(2) A-iii ; B-i ; C-iv ; D-ii
(3) A-iii ; B-i ; C-ii ; D-iv
(4) A-i ; B-ii ; C-iii ; D-iv

85) Match the column I with column II.

Column - I		**Column - II**	
A.	*Vallisneria*	(i)	Insect pollination
B.	Water hyacinth	(ii)	Water pollination

C.	*Yucca*	(iii)	Tallest flower
D.	*Amorphophallus*	(iv)	Moth pollination

Correct answer is –

(1) A – i, B – ii, C – iii, D – iv

(2) A – ii, B – i, C – iv, D – iii

(3) A – iv, B – iii, C – i, D – ii

(4) A – iii, B – i, C – ii, D – iv

86) Match the columns I & II and choose the correct combinations.

	Column-I		**Column-II**
(a)	Funicle	(i)	Small opening of ovule
(b)	Integuments	(ii)	Stalk of ovule
(c)	Chalaza	(iii)	Protective envelopes of ovule
(d)	Hilum	(iv)	Junction part of ovule and stalk
(e)	Micropyle	(v)	Base of ovule

(1) a-ii, b-iii, c-v, d-iv, e-i

(2) a-iii, b-ii, c-i, d-iv, e-v

(3) a-ii, b-iii, c-v, d-i, e-iv

(4) a-v, b-iv, c-iii, d-ii, e-i

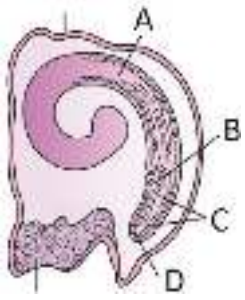
87) **Statement-1:** An ovule generally has a single embryosac formed from a single megaspore.
Statement-2: Ovules generally differentiate a single megaspore mother cell in the micropylar region of the nucellus.

(1) Both statement 1 and statement 2 are correct

(2) Statement 1 is correct but statement 2 is incorrect

(3) Statement 1 is incorrect but statement 2 is correct

(4) Both statement 1 and statement 2 are incorrect

88) 

Label the given diagram.

(1) A - Cotyledon, B - Shoot apical meristem,
C - Hypocotyl root axis, D - Root tip

(2) A - Cotyledon, B - Root apical meristem,
C - epicotyl root axis, D - Root tip

(3) A - Endosperm, B - Shoot apical meristem,
C - Hypocotyl root axis, D - Root tip

(4) A - Cotyledon, B - Shoot apical meristem,
C - epicotyl shoot axis, D - Root tip

89) Given below are two statements :
Statement-I : Most fruits develop only from the ovary and are called true fruits. In a few such as apple, strawberry etc, the thalamus also contributes to fruit formation, such fruits are called false fruits.
Statement-II : Non-albuminous seeds have no residual endosperm as it is completely consumed during embryo development (eg. pea, groundnut). Albuminous seeds retain a part of endosperm as it is not completely used up during embryo development (eg. wheat, maize, barley, castor).
In the light of above statements, choose correct answer from the options given below :

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct but Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

90) Read the following statements and mark them as True (T) or False (F).
a. Coconut water from tender coconut has thousands of free nuclei.
b. Both endosperm and perisperm are similar in ploidy level and their origin.
c. Coleoptile is a hollow foliar structure that encloses leaf primordia in embryo.
d. Single, large shield shaped cotyledon of wheat embryo is known as epiblast.

	a	**b**	**c**	**d**
(1)	T	F	F	F
(2)	T	F	T	F
(3)	T	F	F	T
(4)	F	T	F	F

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	1	4	4	3	2	1	4	4	2	1	3	2	4	4	2	2	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	4	2	2	1	2	3	2	4	2	3	3	1	1	4	1	2	4	3	3
Q.	41	42	43	44	45															
A.	1	4	2	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	4	2	4	3	4	3	1	1	2	3	2	4	4	3	4	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	1	2	1	2	3	1	4	3	2	2	1	3	4	3	3	2	1	2	2
Q.	86	87	88	89	90															
A.	1	1	3	3	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	4	1	2	4	1	1	2	4	1	2	4	4	1	4	3	1	2	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	3	4	3	3	2	3	1	3	1	4	4	1	4	1	2	3	2	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	2	4	2	2	4	2	2	1	3	3	1	3	3	1	3	4	2	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	3	2	3	2	3	2	1	4	4	4	3	3	2	1	2	4	4	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	4	2	2	1	1	1	1	2										

SOLUTIONS

PHYSICS

1) $y = 2\sin\omega t + 2\cos\omega t + 4\sqrt{2}\sin(\omega t + \pi/4)$

$= 2\sqrt{2}\sin(\omega t + \pi/4) + 4\sqrt{2}\sin(\omega t + \pi/4)$

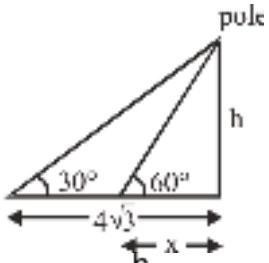
$= 6\sqrt{2}\sin(\omega t + \pi/4)$

2) $I = \int_1^2 \frac{dx}{(3x+4)} = \left[\frac{\log_e(3x+4)}{3}\right]_1^2$

$= \frac{1}{3}\left[\log_e(3\times 2+4) - \log[3\times 1+4]\right]$

$= \frac{1}{3}[\log_e 10 - \log_e 7] = \frac{1}{3}\log_e\left(\frac{10}{7}\right)$

3) $\tan 30° = \frac{h}{4\sqrt{3}} \Rightarrow h = 4m$



$\tan 60° = \frac{h}{x} \Rightarrow \sqrt{3} = \frac{4}{x} \Rightarrow x = \frac{4}{\sqrt{3}}m$

4) Shape of graph is like cosθ. If it is y = A cos kx here peak value is 1 so A = 1 and cycle completes at $x = \frac{2\pi}{3}$ so $k\left(\frac{2\pi}{3}\right) = 2\pi \Rightarrow K = 3$
so y = cos 3x

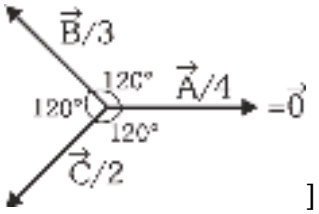
5) $\theta = \frac{arc}{Radius}$

$R = \frac{arc}{\theta} = \frac{\frac{720}{\pi}\times 10^6}{\frac{\pi}{180}\times\frac{3}{2}} = \frac{8640000\times 10^6}{10}$

$= 8.64 \times 10^9$ m

6) $(\vec{P}\times\vec{Q}).\vec{R} = 0$ (coplanar).

7) $\frac{3\vec{A}}{4}+\frac{\vec{B}}{3}+\frac{\vec{C}}{2}=\frac{\vec{A}}{2}+\left(\frac{\vec{A}}{4}+\frac{\vec{B}}{3}+\frac{\vec{C}}{2}\right)=\frac{\vec{A}}{2}+\vec{0}=\frac{\vec{A}}{2}$



[ ]

8)

Question Asking About:
Given: Column-I contains vector conditions .Column -II contains corresponding result .
To determine: Correct matching between Column -I and Column-II
Key Concept:
Vector Magnitude and Resultant Conditions
Formula
$-\frac{1}{2}\Rightarrow\theta=120^{\circ}$ • If $|A|=|B|$ and $|A+B|=|A|$, then cosθ =
• Resultant of two forces:
$R_{max}=F_1+F_2, R_{min}=—F_1-F_2|$
⇒ R = 12N is possible
• Angle between $A=2\hat{i}+2\hat{j}$ and $B=2\hat{k}$ is 90° since they perpendicular.
• Magnitude of resultant:
$R=\sqrt{(2)^2+(1)^2+(3)^2}=\sqrt{14}$
Solution/Explanation/Calculation:
(A) → q(120°)
(B) → r (12N)
(C) → p (90°)
(D) → s$\sqrt{14}$
Conclusion:
Hence, option (1) is correct.

9) According to definition of cross–product $\vec{C}+\vec{D}$ is
perpendicular to both .
i.e., $\vec{A}.(\vec{C}+\vec{D})=0$
or $\vec{A}.\vec{C}+\vec{A}.\vec{D}=0$
or A(component of $\vec{C}$ along $\vec{A}$) +A (component of $\vec{D}$ along $\vec{A}$) = 0
or component of $\vec{C}$ along $\vec{A}$ = – component of $\vec{D}$ along $\vec{A}$

10) absinθ = $\frac{1}{\sqrt{3}}$ab cosθ
tanθ = $\frac{1}{\sqrt{3}}$
θ = 30°
$\left|\vec{a}+\vec{b}\right|=\sqrt{a^2+b^2+2ab\ \cos 30^{o}}$

$$\left|\vec{a}+\vec{b}\right| = \sqrt{a^2+b^2+\sqrt{3}ab}$$

11) $\left[50\hat{i}+\left(50\sqrt{3}-10t\right)\hat{j}\right]\cdot\left[-50\hat{i}+\left(50\sqrt{3}-10t\right)\hat{j}\right]=0$

$-2500+\left(50\sqrt{3}-10t\right)^2=0$

on solving

$t=5\left(\sqrt{3}-1\right)$

12) Horizontal component of velocity should be remain same.

13) $\dfrac{d\sqrt{v}}{dt}=\dfrac{1}{2\sqrt{v}}\dfrac{dv}{dt}=\tan\theta$

$\Rightarrow a = 2\sqrt{v}\tan\theta$

$= 2 \times 3 \times \tan 60$

14) **Question Explanation:** Calculate the time taken for two trains moving in opposite directions to completely cross each other, given their lengths, initial speeds, and accelerations.
Concept: Relative motion in one dimension.
Solution:
When two objects move relative to each other, it is often simpler to fix one object and observe the motion of the other. This is the concept of **Relative Velocity** and **Relative Acceleration**. For two trains to "pass each other," the total distance the "moving" train must cover relative to the "fixed" train is the sum of their lengths.

Relative Distance (s_{rel}):
To pass completely, the relative distance to be covered is the sum of the lengths of both trains.
$s_{rel} = L_1 + L_2 = 100\ m + 125\ m = 225\ m$
Relative Initial Velocity (u_{rel}):
Since they are moving in opposite directions, their relative speed is the sum of their individual speeds.
$u_{rel} = u_1 + u_2 = 10\ m/s + 10\ m/s = 20\ m/s$
Relative Acceleration (a_{rel}):
Similarly, because they are moving in opposite directions, the magnitude of their relative acceleration is the sum of their individual accelerations.
$a_{rel} = a_1 + a_2 = 0.3\ m/s^2 + 0.2\ m/s^2 = 0.5\ m/s^2$
Finding Time (t):
Using the second equation of motion: $s_{rel} = u_{rel}t + \frac{1}{2}a_{rel}t^2$
$225 = 20t + \frac{1}{2}(0.5)t^2$
$225 = 20t + 0.25t^2$
Multiply the entire equation by 4 to simplify:
$900 = 80t + t^2 \Rightarrow t^2 + 80t - 900 = 0$
Solving the Quadratic Equation:
Factorizing the equation:

(t + 90)(t - 10) = 0
Since time cannot be negative, t = 10 s.
Final Answer: Option (2)

15)

 v = u + at v = 0 + na = na

$$s = \frac{1}{2}an^2 - \frac{1}{2}a(n-2)^2 = \frac{1}{2}an^2 - \frac{1}{2}a[n^2 + 4 - 4n]$$

$$s = \frac{1}{2}an^2 - \frac{1}{2}an^2 - 2a + 2an = 2a(n-1)$$

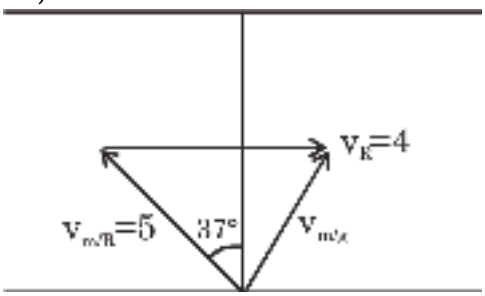
$$\therefore s = \frac{2v(n-1)}{n}$$

16) $\tan 60^\circ = \frac{V_r}{V}$

and $\tan 45° = \frac{V_r}{(1+\beta)V}$

By solving $\beta = 0.73$

17) $\bar{v}_{m/y} = \hat{i} + 4\hat{j}$



$v_y = 4$

$v_x = 1$

$$t = \frac{6}{4} = \frac{3}{2}hr$$

$$D = v_x \times t = \frac{3}{2}km$$

18) $v = \sqrt{169 - 8x}$

$v^2 = 169 - 8x$

Comparing with $v^2 = u^2 + 2as$

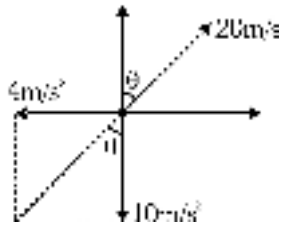
$u = 13$ m/s and $a = -4$ m/s^2

19) $$t = \left(\frac{d}{V_B + V_R}\right) + \frac{d}{(V_B - V_R)}$$

$$= \frac{d}{10} + \frac{d}{5}$$

$3 = \frac{3d}{10} \Rightarrow d = 10$ km

Total distance travelled = 10 + 10 = 20 km

20) 

The ball returns back to boy's hand only if the path of the ball is a straight line. Hence initial velocity and net acceleration must be along the same line. Hence

$\tan\theta = \frac{4}{10} \Rightarrow \theta = \tan^{-1}0.4$

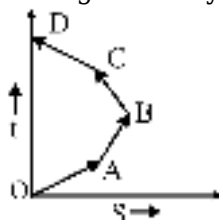
21) **Explain the question :**

We need to interpret the displacement-time graph shown, where motion is forward first and then reversed back, to decide the correct statement.

Concept : This question is based on Average velocity and speed.

Solution :

Average velocity = net displacement / total time.

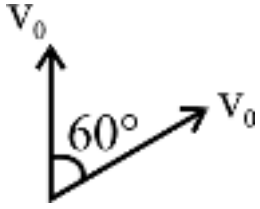


The body returns to the starting point after moving forward, so net displacement is zero, hence average velocity is zero.

Final answer : 3

22)

Final separation
= Initial separation – |Relative displacement|
= 300 – [(1/2) × 40 × 10 + (1/2) × 20 × 8]
= 20 m

23) 

$v_{rel} = v_0$

$d = v_{rel \times t}$

24)

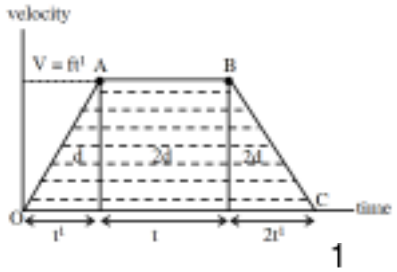
From 0 to A

$V = 0 + ft^1 = ft^1$

From B to C

$$0 = ft^1 - \frac{f}{2} \times t^{11}$$

$\Rightarrow t^{11} = 2t^1$



distance from 0 to A $d = \frac{1}{2} \times t^1 \times V$

distance from B to C $d' = \frac{1}{2} \times 2t^1 \times V$

$d^1 = 2d$

So, distance from A to B

$= 5d - d - 2d = 2d$

So, $2d = ft^1 t \Rightarrow t^1 = \frac{2d}{ft}$

$$d = \frac{1}{2} \times \left(\frac{2d}{ft}\right)^2 \times f$$

$$\Rightarrow d = \frac{1}{2} ft^2$$

25) Given equation $y = \sqrt{3}x\left(1 - \frac{x}{5\sqrt{3}}\right)$

Equation trajectory $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

By comparition

(1) $\tan\theta = \sqrt{3} \Rightarrow \theta = 60^\circ = \frac{\pi}{3}\text{rad}$

(2) $R = 5\sqrt{3}$

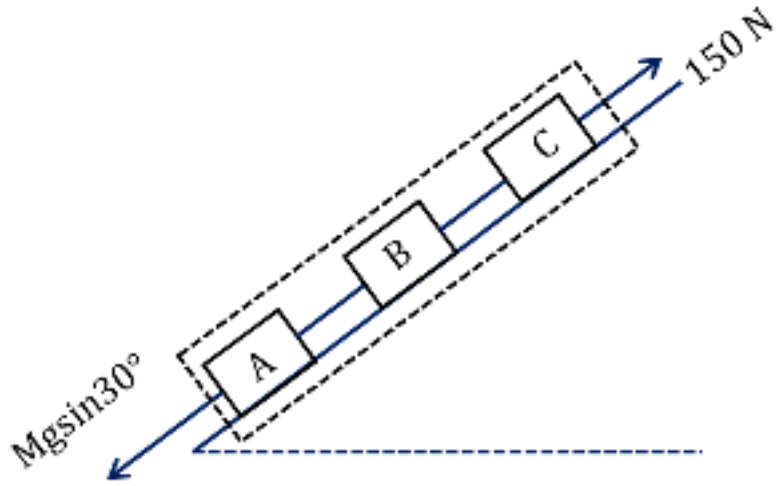
(3) $\tan\theta = \frac{4H}{R} \Rightarrow H = \frac{R}{4}\tan\theta = \frac{5\sqrt{3}}{4} \times \sqrt{3} = \frac{15}{4}$

(4) $H_{max} = \frac{u^2 \sin^2 60^\circ}{2g} \Rightarrow \frac{15}{4} = \frac{u^2}{20} \cdot \frac{3}{4} \Rightarrow$ u = 10 m/s

26)

Considering all three blocks together as system

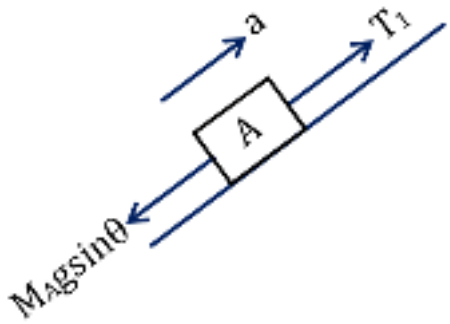
Where M = 2 + 3 + 5 = 10 Kg



$F_{net} = Ma$
$150 - Mg\sin30° = 10\ a$

$$a = \frac{150 - 100 \times \frac{1}{2}}{10} = 10\ m/s^2$$

Now considering block A as system
$T_1 = mg\sin30° = M_A a$

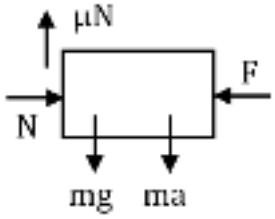


$T_1 - M_A g\sin30° = 2 \times a$

$$T_1 = 2 \times 10 + 2 \times 10 \times \frac{1}{2}$$

$T_1 = 30\ N$

27)



for equilibrium of block with respect to lift
$N = F$
$\mu N = mg + ma$
$0.5F = 10 \times 10 + 10 \times 2$
$F = 120 \times 2$
$F = 240\ N$

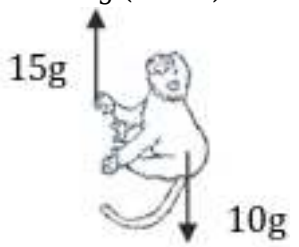
28) P will increase till t = 5, then P decrease
P is maximum at t = 5

$$\int Fdt = \Delta P$$

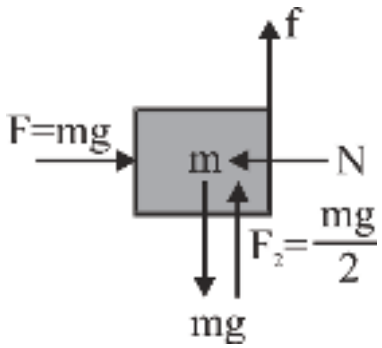
$10 \times 5 = m (v_f - v_i)$
$50 = 2 (v - 2)$
$v = 27$ m/s → maximum speed

29) As there is no tendency of relative slipping between the block and cube, the friction force is zero.

30) To left the block off the ground T = 15 g (N = 0)



Applying NLM on monkey, we have $15 g - 10 g = 10 a$
$\Rightarrow a = g/2$

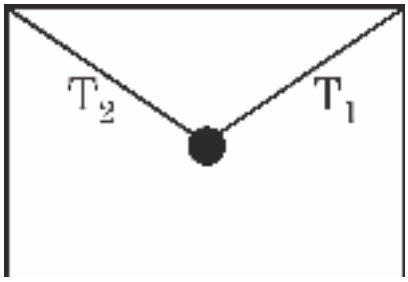


31)
Here net driving force
$= mg - \frac{mg}{2} = \frac{mg}{2}$ downward
Hence friction will act upward and its magnitude should be $f = \frac{mg}{2}$. If the block 'm' is stationary the friction between m and the wall should be static.
$f \le f_{lim}$

$$\frac{mg}{2} \leqslant \mu . N \Rightarrow \frac{mg}{2} \le \mu mg \Rightarrow \mu \ge \frac{1}{2}$$

32) $kx\cos 53 = mg$

$$a = \frac{kx \sin 53}{m} = g \tan 53 = \frac{4g}{3}$$

33) 

$T_1\sin30° + T_2\sin30° = mg$

$2mg \times \frac{1}{2} + T_2\frac{1}{2} = mg$

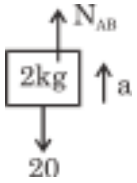
$T_2 = 0$

$T_1\cos30° = ma.$

$2m g \times \frac{\sqrt{3}}{2} = m a$

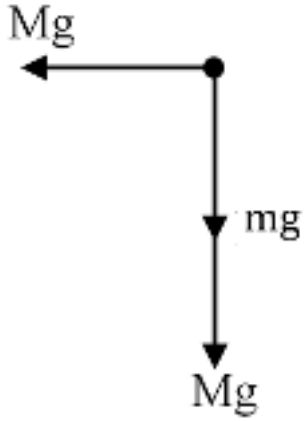
$a = \sqrt{3}g$

34) FBD



$N_{AB} = 2 \times 2 + 20$

$N_{AB} = 24N$



35)

Force on pulley $= \sqrt{(Mg)^2 + (M+m)^2 g^2}$

$= g\sqrt{M^2 + (M+m)^2}$

36) Pseudo force Fp = 30N

f = 20 N

$a_{rel} = \frac{30-20}{10} = 1 m/s^2$

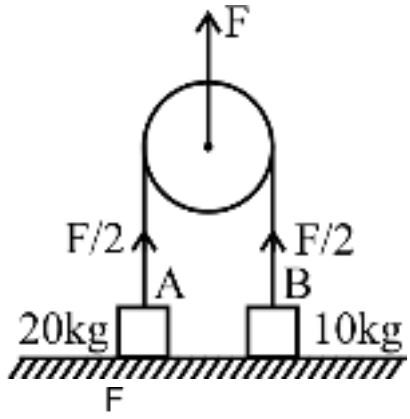
$s_{rel} = \frac{1}{2}a_{rel}t^2$

$s = \frac{1}{2}(1)t^2$

$t^2 = 10 \Rightarrow t = \sqrt{10}s$

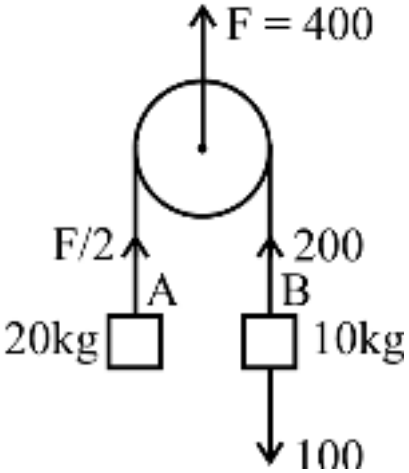
ट्राली द्वारा तय की गई दूरी $S_T = \frac{1}{2} \times 3 \times 10 = 15m$

37)



$F = \frac{F}{m}$

$F = 400 \Rightarrow$



$F_{net} = 200 - 100 = 10 \times a$

$a = 10$ m/sec

38)

$a = 2m/s^2(\downarrow)$

$60 - T = 12$

$T = 48$ N

39) Torque about

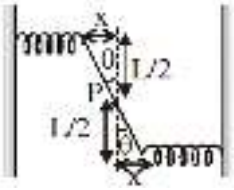
$P = -(kx)\ \frac{L}{2} + \left(-kx\frac{L}{2}\right) = -kxL = -k\frac{L^2}{2}\theta$

For small angle θ, $x = \frac{L}{2}\theta$; $\tau = -I\alpha$

$\Rightarrow -\frac{KL^2}{2}\theta = \frac{ML^2}{12}\alpha$

$\Rightarrow \frac{-6K\theta}{M} = \alpha$

$\Rightarrow \omega = \sqrt{\frac{6K}{M}}$ and $f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{6K}{M}}$



40) $\frac{1}{2}kx_0^2 = \frac{1}{2}\frac{mM}{m+M}\omega^2 x_0^2$

$\omega = \sqrt{\frac{k(m+M)}{mM}} = \frac{2\pi}{T}$

$T = 2\pi\sqrt{\frac{mM}{(m+M)k}}$

41) $y = 1 \sin \omega t + \sqrt{3}\cos\omega t$

$y = 2\sin(\omega t + \frac{\pi}{3})$

For breaking off plank

$\omega_{min} = \sqrt{\frac{g}{a}} = \sqrt{\frac{g}{2}}$

At $t = 0$, $\phi^0 = \frac{\pi}{3}$, It leaves when $\phi = \frac{\pi}{2}$

$\Delta\phi = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$

$t = \frac{T}{12} = \frac{1}{12} \times 2\pi\sqrt{\frac{2}{g}} = \frac{\sqrt{2}}{6}\frac{\pi}{\sqrt{g}} = \frac{\pi}{6}\sqrt{\frac{2}{g}}$

42) When the mass m is attached, the new equilibrium extension (x_e) is given by

$(k \times x_e = m \times g)$

$\Rightarrow (x_e = (m \times g)/k)$

Initially, the spring was stretched by (d). The difference between the initial position and equilibrium position is $(A = |x_e - d|)$

This distance becomes the amplitude of oscillation when the mass is released.

Hence,

Amplitude $(A) = |(m \times g)/k - d|$

43) $y = a\sin(\omega t + \phi)$

$y = 2 \sin(\omega t + \phi)$

at $t = 0$, $y = -2$

$-2 = 2 \sin\phi \Rightarrow f = -\frac{\pi}{2}$

$\Rightarrow y = 2 \sin\left(\frac{10t}{3} - \frac{\pi}{2}\right)$

44) When a little mercury is drained off, the position of c.g. of ball falls (w.r.t fixed end) so that

effective length of pendulum increases hence T increases.

45) Velocity of a particle executing S.H.M. is given by

$$v = \omega\sqrt{a^2 - x^2} = \frac{2\pi}{T}\sqrt{A^2 - \frac{A^2}{4}}$$

$$= \frac{2\pi}{T}\sqrt{\frac{3A^2}{4}} = \frac{\pi A\sqrt{3}}{T}$$

CHEMISTRY

46)

Explanation -

The question asks for the correct order of increasing acid strengths for the given series of chlorine oxyacids.

Concept - oxy acids

Solutions -

The given acids are oxyacids of chlorine:

A. HOCl : Oxidation state of Cl = +1

B. HOClO : Oxidation state of Cl = +3

C. $HOClO_2$: Oxidation state of Cl = +5

D. $HOClO_3$: Oxidation state of Cl = +7

Therefore, the increasing order of oxidation states of chlorine in these acids is +1 < +3 < +5 < +7. Following the trend, the increasing order of acid strength should be:

HOCl < HOClO < $HOClO_2$ < $HOClO_3$

Final Answer - Option (2)

47) **Explanation :** Amphoteric hydroxide is asked

Solution/Explanation/Calculation :
$Al(OH)_3$ & $Zn(OH)_2$ are amphoteric in nature
$Ba(OH)_2$ ⇒ basic hydroxide

Hence, the correct answer is option (3).

48)

group 16 is also known as chalcogens or oxygen family

49) $\phi = \frac{\text{charge}}{\text{size}}$

50)

Acidic strength of nonmetallic hydrides ∝ B.L. basic nature of metallic hydroxide ∝ size

51) **Question Explanation:** Match the atomic number of elements with their periodic block.
Concept: Periodic Blocks
Solution: Z=37 (Rubidium) is [Kr] $5s^1$, an s-block element. Z=42 (Molybdenum) is [Kr] $4d^5 5s^1$, a d-block element. Z=34 (Selenium) is [Ar] $3d^{10} 4s^2 4p^4$, a p-block element. Z=92 (Uranium) is [Rn] $5f^3 6d^1 7s^2$, an f-block element.
Final Answer: Option (3)

52)

$(n-1)d^{1-10}ns^{1-2}$ (n = 3)
2d is not possible.

53)

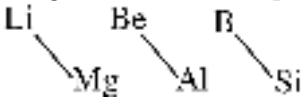
Na = [Ne]$3s^1$ (s-block)
one p-orbital holds $2e^-$

54)

X^{+2}[Ar]$3d^8$ = 26 'e'
hence X^0 = 28 'e'
Y^{+3}[Ar]$3d^3$ = 21 'e'
hence Y^0 = 24 'e'

55)

Diagonal relationship



56)

Metallic oxides (Bi_2O_3) → Basic
Nonmetallic oxides (Cl_2O_7) → Acidic
Neutral oxide → N_2O, NO, H_2O, CO
Amphoteric → ZnO, BeO, Al_2O_3, PbO/PbO_2, SnO/SnO_2

57) **Question Explanation:** Identify where Xenon has the highest electronegativity among the options.
Concept: Oxidation State & Electronegativity
Solution: Electronegativity increases as the positive oxidation state of an atom increases. Xenon has its highest oxidation state (+6) in XeF_6, making it the most electronegative in this compound.
Final Answer: Option (3)

58)

Highest EA in periodic table = Cl

$$Cl > \underset{F^1}{O} > \underset{F^3}{N} > C$$

So (stable)

59)

$Li^+ \geq Mg^{+2}$ (Ionic radius)
diagonal relationship

60)

$Z_{eff} \propto$ I.P.

Shell No. $\propto \frac{1}{IP}$

$$\underset{\text{Same period}}{Na < S < P} < \underset{2^{nd}\text{ period smallest}}{F}$$

So $Z_{eff} \uparrow$ IP $\uparrow$ almost Na, S & P

61)

IP of neutral atom > EA of neutral atom
So x > y

62)

$S^- \rightarrow S^{-2}$ 2nd EA, Endo
$N \rightarrow N^-$ 1st EA endo
(half filled)

$Al^{-2} \xrightarrow{1^{st}\ IP} Al^{-1}$ endo

$$Na^+_{(g)} + Cl^-_{(g)} \xrightarrow[\text{Lattice energy}]{} NaCl_{(s)}$$

63)

$Al_{(g)} \rightarrow Al^{+}_{(g)}$

64)

2nd IP

K^{+} $|Ar|$ $>$ Cu^{+} $3d^{10}$

65)

Neutral do not react with acid as well as base
So $N_2O + NaOH \rightarrow$ NO Reaction
$N_2O + HCl \rightarrow$ NO Reaction

66) $-\frac{1}{2}\frac{d[N_2O_4]}{dt} = \frac{1}{4}\frac{d[NO_2]}{dt}$
Given:
$\frac{d[N_2O_4]}{dt} = -k_1\frac{d[NO_2]}{dt} = k'$
Substitute:
$-\frac{1}{2}(-k) = \frac{1}{4}k'$
$\frac{k}{2} = \frac{k'}{4}$
k' = 2k

67)

Step 1 : Compare Exp. 1 and 2
([B] constant [A] doubles from 0.1 to 0.2 M\(\))

$$\frac{r_2}{r_1} = \frac{8.4 \times 10^{-3}}{2.1 \times 10^{-3}} = 4$$

$$\left(\frac{[A]_2}{[A]_1}\right)^m = 2^m = 4 \Rightarrow m = 2$$

Step 2 : compare exp. 2 and 3
([A] constant [B] doubles from 1.0 to 2.0 M)

$$\frac{r_3}{r_2} = \frac{8.4 \times 10^{-3}}{8.4 \times 10^{-3}} = 1$$

$$\left(\frac{[B]_3}{[B]_2}\right)^n = 2^n = 1 \Rightarrow n = 0$$

68) **Explain the question :** Find the rate law for the given mechanism.
Concept : This question is based on rate law from mechanism (fast equilibrium + slow step)
Solution :
Fast equilibrium :

$$\frac{K_1}{K_2} = \frac{[O_2][O]}{[O_3]} \Rightarrow [O] = \frac{K_1}{K_2}\frac{[O_3]}{[O_2]}$$

Slow (rate-determining) step :

$r = k_3 [O_3][O]$

Substitute [O]:

$$r = k_3[O_3]\left(K\frac{[O_3]}{[O_2]}\right) = \left|\frac{K_2 \times K_1}{K_2}\right| [O_3]^2[O_2]^{-1}$$

Final answer : 2

69)

For simple reaction, order is equal to molecularity. There fore the statement "order is not equal to the molecularity" is incorrect.

70)

For zero order reaction

$[A]_t = [A]_0 - kt$

slope = $-k = -0.48$

$\Rightarrow k = 0.48$

71)

$$k\,t = 2.303\ \log\left(\frac{q_0}{q_t}\right)$$

$$\Rightarrow \frac{t_{80\%}}{t_{90\%}} = \frac{\log\left(\frac{100}{20}\right)}{\log\left(\frac{100}{10}\right)}$$

$$\Rightarrow \frac{70}{t_{90\%}} = \frac{\log(5)}{\log(10)} \Rightarrow \boxed{t_{90\%} = 100\ \text{min.}}$$

72)

($t_{1/2} = 0.693/k$), independent of $[A]_0$ → gives a horizontal line (i).

• log(At) vs t gives a straight line with negative slope (–k/2.303) (iii).

• (ii) is incorrect because t1/2 does not decrease with $[A]_0$.

73)

Species	Initial Pressure	Change	Final Pressure
A	P_0	$-P_0\alpha$	$P_0(1-\alpha)$
B	0	$+3P_0\alpha$	$3P_0\alpha$
Total (P_r)	-	-	$P_r = P_0(1+2\alpha)$

From total pressure relation: $\alpha = \frac{P_T/P_0 - 1}{2}$

For a first-order reaction: $kt = \text{In}\left(\frac{P_0}{P_A}\right)$

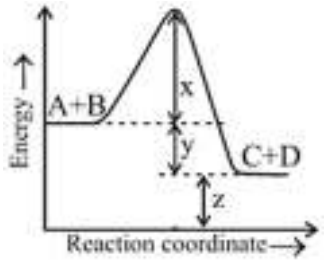
Substituting $P_A = P^0(1-\alpha) = \frac{3P_0 - P_r}{2}$:

$k = \frac{1}{t}\text{In}\left(\frac{P_0}{P_0}\right) = \frac{1}{t}\text{In}\left(\frac{2P_0}{3P_0 - P_T}\right)$

Final answer: option (4)

74)

Forward reaction (A + B → C + D) : since $E_{product}$ is lower than $E_{reactants}$ by y, the forward reaction
is exothermic.
$\Delta H_{fwd} = -y$
Reverse reaction (C + D → A + B) : The reverse reaction is endothermic and has the opposite sing.
$\Delta H_{rev} = -\Delta H_{fwd} = -(-y) = +y$
Activation energy (E_a) for the reverse reaction :
The reactants for the reverse reaction are C + D.
$E_{a,rev}$ is the energy difference between the transition state and the reactants C + D.
From the diagram, the energy from the transition state to A + B is x. The energy from A + B to C + D is y.
Therefore, $E_{a,rev}$ is the sum of these two vertical segments : x + y.



Final answer : 3

75)

$\log k = \log A - \frac{E_a}{RT \times 2.303}$

$\log k = 6 - \frac{1000}{2T}$ (given reaction)

$\frac{E_a}{2.303 \times R} = \frac{1000}{2}$

$E_a = \frac{1000}{2} \times 2.303 \times \frac{8.314}{1000}\ \frac{kJ}{mole}$

$E_a = 9.57\ kJ/mole$

76) **Solution:** Catalyst changes reaction rate by lowering activation energy but does not alter initial or final energy states. Hence heat of reaction remains unchanged, while rate increases. Reason is true but not explanation.

77)

(1) k is independent of concentrations of A and B at a fixed temperature.
→ **True (T)**

(2) Rate = k[A][B]

Units of k = (mol L^{-1} s^{-1})/(mol L^{-1} × mol L^{-1})
= L mol^{-1} s^{-1}

Not s^{-1}.
→ **False (F)**

(3) Overall order = 1 + 1 = 2

For a second-order reaction, half-life depends on initial concentration and is not constant.
→ **False (F)**

(4) For 2A + B → C,

$-(1/2)d[A]/dt = d[C]/dt$

$\Rightarrow -d[A]/dt = 2\ d[C]/dt$

So, rate of formation of C is **half** the rate of consumption of A.
→ **False (F)**

78) V.P. of pure liquid depends only on temperature.

79) 88ºC (B.P.) V.P. of solution = 1 atm = 760 mm
$760 = P°_A X_A + P°_B (1 - X_A)$
$760 = 960\ X_A + 380(1 - X_A)$
$X_A = 0.655$

80) Max. boiling azeotropes form by negative deviation from Raoult's law.

81) $\frac{P^o - P^S}{P^S} = \frac{n_B}{n_A}$ or $\frac{P^o - P^S}{P^o} = X_B$

$$\frac{100-99}{99} = \frac{\left(\frac{4}{M}\right)}{\left(\frac{54}{18}\right)}$$

M = 132

82) $$\frac{1}{P_T} = \frac{Y_A}{P^o_A} + \frac{Y_B}{P^o_B}$$

$$= \frac{0.75}{75} + \frac{0.25}{25}$$

$$\frac{1}{P_T} = \frac{1}{100} + \frac{1}{100} = \frac{2}{100} = \frac{1}{50}$$
$P_T = 50$

83) $P_A = Y_A P_T$, $P_T = P_A + P_B = 2.8 + 4.2 = 7$
$2.8 = Y_A\ 7$
$Y_A = \frac{2.8}{7} = 0.4$

84) V.P. ↓ solute ↑ solvent ↓ in S_2
(Osmosis occur towards higher solvent to lower solvent amount.

85) Osmosis occur towards higher solvent to lower solvent amount.

86) $K_b = \frac{R(T_b^o)^2 M_A}{1000 \Delta H_v}$

87) $\Delta T_f = K_f\ m$
$(327\text{-}322) = \frac{8.5\left(\frac{W}{120}\right)}{425} \times 1000$

88) Osmotic pressure and R.L.V.P. depends on relative amount of solute.

89) (i = 2) given
$\alpha = \frac{i-1}{(n-1)} = \frac{1}{(n-1)}$
A → n = 5, α = 0.25
B → n = 2, α = 1
C → n = 3, α = 0.5
D → n = 4, α = 0.33

90) $\Delta T_f = \frac{(2a+3b)}{1} \times K_f$
= (2a + 3b) × 1.86
(For min ΔT_f mole of only KCl are present is 0.1 mole)
= 2 × (0.1) × 1.86 = 0.372°C
For max. ΔT_f moles of $BaCl_2$ = 0.1
= 3 × (0.1) × 1.86 = 0.558°C
Range is 0.372°C to 0.558°C

BIOLOGY

91)

NCERT-XIth, Pg. # 24

92)

NCERT-XIth, Pg. # 24

93) NCERT Pg. # 18, last para

94) NCERT-XI Pg # 33, Table

95) **Explain Question :**The question asks to identify the starting point of the free-living gametophyte stage in the life cycle of liverworts.
Concept : This question is based on life cycle of liverworts
Explanation : Spores produced in the capsule germinate to form free-living gametophytes.
Final answer : 2

96)

Explain Question : The question asks for the name of the first structure that a moss spore develops into immediately after germination.
Concept : This question is based on mosses.
Explanation : The spores on germination develop into protonema, which is the first stage in the life cycle of a moss.
Final answer : 4

97) **Explain Question :** The question has provided an assertion-reason question. I need to evaluate the truthfulness of both statements and determine if the reason provides a correct explanation for the assertion.
Concept : This question is based on Rhodophyceae
Explanation :
Assertion - The members of rhodophyceae are commonly called red algae. - correct Rhodophyceae are called red algae.
Reason - Red algae have predominance of the red pigment, r-phycoerythrin in their body. - correct r-phycoerythrin is the red pigment dominant in these organisms.
Does the Reason explain the Assertion?
Yes. The name "red algae" comes because of the red pigmen
Final answer : 1

98) **Explain Question :**The question presents an Assertion (A) and a Reason (R) about the bryophyte life cycle asks for an analysis of their relationship
Concept : This question is based on Bryophyta, General Characters And Life Cycle
Solution :Assertion: In bryophytes, zygote produces a multicellular body called a sporophyte. Correct – This is a defining feature of the bryophyte life cycle. Reason: Zygote does not undergo reduction division immediately. Correct – The zygote first divides mitotically to form the sporophyte. The reason directly explains why the zygote forms a multicellular sporophyte instead of forming spores immediately.

Final answer : 1

99)

Explain Question : The question asks you to identify the correct chronological sequence of developmental stages and structures in the life cycle of a moss.
Concept : This question is based on mosses.
Explanation : Zygote → develops into sporophyte →Sporophyte produces spores by meiosis Spores germinate into protonema →Protonema develops buds → develops into mature gametophyte.
Final answer : 2

100) **Explain Question :** The question asks to identify all the incorrect statements about the characteristics of a moss protonema.
Concept : This question is based on mosses.
Explanation :
(b) Non-Green - Protonema is green (false)
(c) Unbranched - Protonema is branched (false)
(e) Develops from gamete - Protonema develops from spore, not gamete (false)
Final answer : 4

101)

Explain Question : The question provides several morphological characteristics of a plant and asks for its identity from a list of options.
Concept : This question is based on lower gymnosperms.
Explanation : "Cycas is a palm-like gymnosperm with an unbranched stem, persistent woody leaf bases, and pinnately compound leaves with sessile leaflets having a midrib but no lateral veins."
Final answer : 1

102)

NCERT-XIth, Pg. # 32

103)

NCERT-XIth, Pg. # 28-33

104) NCERT (XI) Page # 18

105) NCERT XI Pg. # 36, 38

106) **Explain Question :** The question asks for features that help reduce transpiration in conifers.
Concept : This question is based on gymnosperm, general characters and life cycle.

Explanation : "Conifers have needle-like leaves with thick, waxy cuticle and sunken stomata to reduce transpiration."
Final answer : 4

107) **Explanation:** Red Algae (Rhodophyceae): Gametes and spores are non-flagellated and non-motile and Blue-Green Algae (BGA/Cyanobacteria) No flagellated or reproductive stages.
The correct answer is: 3
Red Algae & Blue Green Algae

108)

Explain Question: Question asks about identifying the correct statements regarding Cycas.
Concept: This question is based on Lower Gymnosperm
Solution:
(I)Cycas has coralloid roots containing blue-green algae (Anabaena/Nostoc) → correct.
(II)In Cycas, female cone is absent; female reproductive structures are megasporophylls → correct.
(III)Circinate vernation is characteristic of ferns, In Cycas yong leaf show → correct.
(IV)Gymnosperm xylem mainly has tracheids, not vessels → incorrect.
Final answer: 1

109)

NCERT XI Pg # 30

110) NCERT (XIth) Pg. # 32, Para 1st

111) **Explanation:** The question asks which structure is absent in bacteria, i.e., not a component of a prokaryotic cell.
Concept: Prokaryotic cell structure (Bacteria)
Solution : Mitochondria → Not present in bacteria (they lack membrane-bound organelles). Instead, respiration occurs in the plasma membrane.
Final Answer: Mitochondria

112)

NCERT-XIth, Pg. # 90

113)

NCERT XI Pg # 93

114) **Explain Question :** Which of the following is true for nucleolus.
Concept : This question is based on Nucleus.
Explanation : The nucleolus is a dense, non-membrane bound structure in the nucleus responsible for ribosomal RNA (rRNA) synthesis and ribosome assembly.

Final answer : 3

115) **Question Explanation:** Structure of a cilium in cross-section.
Concept: Axoneme arrangement
Solution: The cross-section of a cilium or flagellum, called the **axoneme**, typically shows a **9 + 2 arrangement** of microtubules. This means there are nine pairs of peripherally arranged doublet microtubules and two central singlet microtubules. There are also nine radial spokes extending from each peripheral doublet to the central sheath.
Final Answer: 3

116)

NCERT-XIth, Pg. # 95

117) **Explain Question :** The question asks to identify which of the given options is not a component of the cell's cytoskeleton
Concept : This question is based on cytoskeleton
Explanation :"The cytoskeleton in a cell is composed of microtubules, microfilaments and intermediate filaments."Microbodies are not a part of the cytoskeleton.
Final answer : 3

118) Miscellaneous

119) **Explain Question:** The question is asking which features are common to mitochondria, chloroplasts, and bacteria.
Concept : This question is based on Mitochondria and chloroplasts, bacteria.
Explanation : Mitochondria and chloroplasts are semi-autonomous organelles and share many features with **bacteria** due to their **endosymbiotic origin**.
Common features:
• **Circular double-stranded DNA** → present in all three.
• **Binary fission** → all divided by binary fission.
• **Porin proteins** → present in outer membranes of mitochondria, chloroplasts, and bacteria.
Final answer (3) : B, C & D only

120)

NCERT-XIth, Pg. # 95-97

121)

NCERT-XIth, Pg. # 98

122) **Explanation** : The question asks which statement about the plasma membrane is not correct.
Concept : plasma membrane Explanation: Plasma membrane is a fluid mosaic of lipids and proteins. Integral proteins are either partially embedded or span the entire membrane, not just

lying on the surface.
Final answer : Option 4: Integral proteins either lie on the surface or totally buried in the membrane

123) NCERT (XI) Pg#129

124)

NCERT-XIth, Pg. # 99

125) **Question Explanation :** Check which of the four statement about the cell cycle/mitosis are incorrect.
Concept : Interphase
Solution :
(A) True – interphase occupies the majority (=95%+) of the cell cycle.
(B) False – DNA replication begins in the nucleus during S, but centrioles duplicate in the cytoplasm (centrosome), not in the nucleolus.
(C) True – mitosis is called equational division because chromosome number is maintained.
(D) True – mitosis usually yields diploid daughters with identical genetic complement.
Final Answer : Option (1)

126) NCERT (XI) Pg. # 122,123,124

127) **Question Explanation:** Identifying the stage with the clearest chromosome view.
Concept: Mitosis and chromosome structure.
Solution: Metaphase is the best stage to observe chromosome structure because at this point, chromosomes are fully condensed and aligned at the cell's equatorial plate, making them highly visible and distinct under a microscope.
Final Answer: Metaphase

128) **Explanation -** In G_1 phase, the cell has 20 chromosomes and 20 pg DNA.
• During S phase, DNA replication occurs, doubling the DNA content to 40 pg, but the chromosome number remains 20 (each chromosome now has two sister chromatids).
• In anaphase, the sister chromatids separate and move to opposite poles, making them individual chromosomes.
So, in an anaphase cell, there are 40 chromosomes (as chromatids are now considered separate chromosomes) and 40 pg DNA (since no DNA replication occurs during mitosis).
Correct answer: 2. 40 chromosomes / 40 pg DNA

129)

NCERT-XIth, Pg. # 126

130) **Explanation :** The question asks to identify the key event that signals the beginning of the diplotene stage.
Concept : This question is based on sub-stage of Prophase I of meiosis.

Solution: "At the beginning of diplotene, the synaptonemal complex begins to dissolve, and homologous chromosomes start to separate but remain attached at chiasmata."
Final answer : 3

131) **Explain the Question :** The question is asking in which phase of the cell cycle the duplication of centrioles occurs. Centrioles are cylindrical cell structures important for spindle formation during mitosis.
Concept : This question based on S-phase
Explanation : Centriole duplication begins at the start of the S-phase of the cell cycle, along with DNA replication. This ensures that two centrosomes are available for spindle formation in the next mitosis.
Final Answer : 2 S-phase

132) NCERT-XIth, Pg. # 121

133)

NCERT-XI, Pg. # 121

134) **Explain Question :** The question is asking during which stage of the cell cycle you can observe nucleoprotein fibers in a loose and indistinct form.

Concept : This Questions based on cell cycle

Explanation : Interphase nucleus → DNA is in the form of loose chromatin → not individually visible chromosomes.

Final answer : 4

135)

NCERT-XIth, Pg. # 123

136)

Module

137)

NCERT Pg. # 9

138)

Algae & Bryophyta are non vascular cryptogames.

139) **Explain Question:** The question asks for the taxonomic category defined as a collection of species showing close resemblance
Concept: Genus
Solution: A collection of species with close resemblance in floral morphology is called a genus.
Final answer: 2

140)

Explain Question: The question asks about the relation which is similar to the relation of Solanaceae and Convolvulaceae with Polymoniales.

Concept: Order/ Family.

Solution:

The relation of Solanaceae and Convolvulaceae with Polymoniales is similar to the relation occuring in : Felidae and Canidae with Carnivora

So, the correct answer is: 1.

Final answer: Option (1)

141)

NCERT Page No. # 30

142) NCERT Page # 29

143)

Question Explanation: The question asks to arrange taxonomic groups in descending order of rank (from highest to lowest).

Concept: Taxonomic Categories & Hierarchy.

Solution: (a) Triticum → Genus (or species Triticum aestivum)
(b) Monocotyledonae → Class
(c) Poaceae → Family
(d) Poales → Order
Descending order of taxonomic rank:
Class → Order → Family → Genus
b → d → c → a

Final answer: (b → d → c → a

144)

NCERT-XIth, Pg. # 12

145)

Explain Question – The question asks which option shows the correct hierarchical sequence of taxonomic categories.

Concept – The Living World → Taxonomic Hierarchy .

Solution –

Correct hierarchy: Species → Genus → Family → Order → Class → Phylum/Division → Kingdom.

Option 3: Correct (Species → Genus → Family → Order matches the hierarchy).

Final answer – Option 3**.**

146)

The Correct answer is : 1. Cell wall

147) **Explain Question:** Question asks about which statement is not related to Mycoplasma.
Concept: This question is based on Mycoplasma
Solution: Statement about lacking cell membrane is incorrect.
Final answer: 3

148)

NCERT-XIth, Pg. # 12

149) **Question Explanation:** Identifying the primary biological process associated with the specialized cell structure known as the heterocyst in Cyanobacteria.
Concept: Function of Heterocysts in Cyanobacteria
Solution: Heterocysts are specialized, thick-walled cells found in filamentous Cyanobacteria (Blue-Green Algae) like *Nostoc* and *Anabaena*. Their primary biological function is:

A. **N2 fixation:** They contain the oxygen-sensitive enzyme **nitrogenase**, which converts atmospheric nitrogen gas (N_2) into ammonia (NH_3). The heterocyst maintains the necessary **anaerobic environment** by lacking Photosystem II, which normally produces oxygen.

While the surrounding vegetative cells perform photosynthesis, the heterocyst itself is responsible for nitrogen fixation. They are not involved in sexual reproduction.
Final Answer: N_2 fixation

150) NCERT (XI) Pg # 22

151) **Question Explanation**: Identifying a fungal class from its features.
Concept: Characteristics of Fungi Classes.
Solution: The key features are the presence of septate and branched mycelium (A), and reproduction mainly by conidia (D). These, along with being saprophytes/parasites (B) and contributing to mineral cycling (C), are the defining characteristics of the class

Deuteromycetes (Imperfect Fungi).
Final Answer: Deuteromycetes

152)

NCERT-XIth, Pg. # 18

153) NCERT-Page # 25

154)

NCERT-XIth, Pg. # 18

155) **Explain Question :**The question asks you to pick the odd ones based on absence of dikaryophase during sexual reproduction.
Concept : Biological Classification → Kingdom Fungi (Phycomycetes, Ascomycetes, Basidiomycetes).
Solution: Dikaryophase is present in: Ascomycetes. Basidiomycetes
Dikaryophase is absent in: Oomycetes (considered under lower fungi) Phycomycetes (lower fungi)
Therefore, the odd ones with respect to absence of dikaryophase are:
Oomycetes (c) and Phycomycetes (d).
Final answer : (3). c, d

156)

Explain Question : This question asks to identify the incorrect statement from the four options provided regarding viruses.

Concept : This question is based on virus.

Solution : "Viruses contain either DNA or RNA as their genetic material but never both."

Final Answer : option (2)

157)

Mycorrhiza is association between plant root & fungi

158)

NCERT-XI, Pg. # 12,13

159)

NCERT-XIth, Pg. # 15

160) NCERT-XI, Pg. # 16

161)

NCERT XII Pg. # 7

162)

NCERT XII Pg. # 5

163)

Module-8 (Target)

164) NCERT XII Pg. # 9

165)

NCERT XII Pg. # 7

166)

NCERT XII Pg. # 10, 11, 18

167)

NCERT XII Pg. # 11

168)

Target module # 8

169)

NCERT XII Pg. # 15

170)

NCERT XII Pg. # 22, 23

171)

NCERT XII Pg. # 14

172)

NCERT XII Pg. No. 7,11,19,22

173) NCERT XII Pg. No. 15

174)

NCERT XII Pg. # 7,11,14,19

175) NCERT XII Pg. # 14,15

176) NCERT XII Pg. # 09

177) NCERT XII Pg. # 9

178)

NCERT XII Pg. # 21

179) NCERT XII Pg.# 20

180) NCERT XII Pg. # 19,20